
Robotics & State Estimation (RiSE) Tutorials

Lecture Notes

Yulin Yang - yangyulin1@gmail.com
Chuchu Chen - chuchu.chen@gwu.edu
Xingxing Zuo - xingxing.zuo@mbzuai.ac.ae

Robotics & State Estimation Tutorials

<https://github.com/yangyulin/rise-tutorial>



Robotics & State Estimation (RiSE) Tutorials
Lecture Notes - RiSE-2026
Last Updated - June 21, 2026

Contents

1	Lecture 1: Foundations — Probability, Linear Algebra, and Linear Systems	1
1.1	Probability	1
1.1.1	Definition	1
1.1.2	Conditional probability	1
1.1.3	Total probability theorem	1
1.1.4	Bayes' theorem	1
1.1.5	Independence	1
1.2	Random Variables	2
1.2.1	Distribution	2
1.2.2	Probability density function (pdf)	2
1.2.3	Example distributions	2
1.2.4	Conditional distribution and density	2
1.2.5	Total probability theorem for pdfs	3
1.2.6	Continuous Bayes' theorem	3
1.3	Expectation, Variance, and Random Vectors	3
1.3.1	Mean	3
1.3.2	Variance	3
1.3.3	A set of random variables	4
1.3.4	Covariance	4
1.3.5	Random vector	4
1.3.6	Gaussian vector pdf	5
1.4	Maximum Likelihood and Maximum a Posteriori Estimation	5
1.4.1	Optimization	6
1.5	Linear Systems	6
1.5.1	Homogeneous solution	6
1.5.2	Properties of the state transition matrix	6
1.5.3	Output sequence and observability	6
1.5.4	Full LTV solution	7
1.6	Homework Problems and Solutions	7
1.6.1	Variance identity	7
1.6.2	Covariance matrix identity	7
1.6.3	Scalar covariance identity	8
1.6.4	Independence versus uncorrelatedness	8
1.6.5	The χ^2 distribution	9
2	Lecture 2: Rotation, Lie Groups, SE(3), SE₂(3), and Quaternions	11
2.1	Preliminaries: useful series	11
2.2	Frames, notation, and transformations	11
2.3	Defining rotation	12
2.3.1	Euler angles	12
2.3.2	Infinitesimal rotation and the skew-symmetric matrix	12
2.4	Axis-angle and Rodrigues' formula	13
2.4.1	Logarithmic map	14
2.5	SO(3) kinematics	14
2.5.1	Worked applications	16
2.6	Groups: SO(3) and SE(3)	16

2.6.1	Matrix Lie groups	17
2.6.2	Why this (manifold) machinery?	17
2.7	SE(3)	17
2.7.1	More operations	19
2.8	SE ₂ (3)	20
2.9	Quaternions	21
3	Lecture 3: Camera Model, Triangulation, Point/Line/Plane	23
3.1	The camera measurement model	23
3.2	The measurement Jacobian	24
3.3	A few important discussions	24
3.4	The full intrinsic calibration model	27
3.5	The extrinsic (camera–IMU) calibration	28
3.6	Point / line / plane representation	28
3.6.1	Point representation	28
3.6.2	Plane representation	29
3.6.3	Line representation	30
	References	32

1 Lecture 1: Foundations — Probability, Linear Algebra, and Linear Systems

1.1 Probability

1.1.1 Definition

For an event A , the probability $P(A)$ is

$$P(A) = \frac{N_A}{N}, \quad (1)$$

where N_A is the number of occurrences of A and N is the number of all possible outcomes.

1.1.2 Conditional probability

The probability of event A given that event B is true is

$$P(A | B) = \frac{P(A, B)}{P(B)}, \quad P(B) \neq 0, \quad (2)$$

which yields the product rule

$$P(A, B) = P(A | B) P(B) \quad (3)$$

$$= P(B | A) P(A). \quad (4)$$

1.1.3 Total probability theorem

If $\{A_1, \dots, A_n\}$ is a partition (mutually exclusive, $P(A_i, A_j) = 0$ for $i \neq j$) and B is an arbitrary event, then

$$P(B) = P(B | A_1)P(A_1) + \dots + P(B | A_n)P(A_n) \quad (5)$$

$$= \sum_i P(B | A_i) P(A_i). \quad (6)$$

1.1.4 Bayes' theorem

$$P(A_i | B) = \frac{P(B | A_i) P(A_i)}{P(B)} \quad (7)$$

$$= \frac{P(B | A_i) P(A_i)}{\sum_i P(B | A_i) P(A_i)}. \quad (8)$$

1.1.5 Independence

With the inclusion–exclusion relation $P(A \cup B) = P(A) + P(B) - P(A \cap B)$, two events A and B are independent when

$$P(A, B) = P(A | B) P(B) \quad (9)$$

$$= P(B | A) P(A) \quad (10)$$

$$= P(A) P(B) \quad (11)$$

$$= P(B) P(A). \quad (12)$$

1.2 Random Variables

A random variable defines a function that quantifies an experiment. For a sample space $S = \{s_1, \dots, s_n\}$, a random variable is a map $x : S \rightarrow \mathbb{R}$ with $x(s_i) = x_i$.

1.2.1 Distribution

The (cumulative) distribution of a random variable x is

$$F_x(x) = P(x \leq x), \quad (13)$$

i.e. the probability that x takes a value lower than or equal to x .

1.2.2 Probability density function (pdf)

$$p(x) = \lim_{\varepsilon \rightarrow 0} \frac{F_x(x) - F_x(x - \varepsilon)}{\varepsilon} \quad (14)$$

$$= \frac{dF_x(x)}{dx} \quad (15)$$

$$= \frac{dP(x \leq x)}{dx}, \quad (16)$$

hence

$$P(x \leq x) = \int_{-\infty}^x p(\xi) d\xi, \quad (17)$$

$$P(x \leq \infty) = \int_{-\infty}^{+\infty} p(\xi) d\xi = 1, \quad (18)$$

and for an interval

$$P(x_1 \leq x < x_2) = \int_{x_1}^{x_2} p(\xi) d\xi. \quad (19)$$

1.2.3 Example distributions

- Gaussian: $x \sim \mathcal{N}(m, \sigma) \iff p(x) = \frac{1}{\sqrt{2\pi} \sigma} e^{-\frac{(x-m)^2}{2\sigma^2}}.$

- Uniform: $p(x) = \begin{cases} 0 & x < x_1 \\ \frac{1}{x_2 - x_1} & x_1 \leq x \leq x_2 \\ 0 & x > x_2 \end{cases}$

1.2.4 Conditional distribution and density

$$P(x \leq x \mid A) = \frac{P(x \leq x, A)}{P(A)}, \quad (20)$$

$$p(x \mid A) = \frac{dP(x \leq x \mid A)}{dx}. \quad (21)$$

1.2.5 Total probability theorem for pdfs

For a mutually exclusive partition $\{A_1, \dots, A_n\}$,

$$p(x) = p(x \mid A_1)P(A_1) + \dots + p(x \mid A_n)P(A_n). \quad (22)$$

1.2.6 Continuous Bayes' theorem

Using the infinitesimal relations $P(x = x) = p(x) \delta x$ and $P(x = x \mid A) = p(x \mid A) \delta x$,

$$P(A \mid x = x) = \frac{P(A, x = x)}{P(x = x)} \quad (23)$$

$$= \frac{P(x = x \mid A) P(A)}{P(x = x)} \quad (24)$$

$$= \frac{p(x \mid A)}{p(x)} P(A), \quad (25)$$

so that $P(A \mid x = x) p(x) = p(x \mid A) P(A)$, which gives the continuous total-probability and Bayes' relations

$$P(A) = \int_{-\infty}^{+\infty} P(A \mid x = x) p(x) dx, \quad (26)$$

$$p(x \mid A) = \frac{P(A \mid x = x)}{P(A)} p(x) \quad (27)$$

$$= \frac{P(A \mid x = x) p(x)}{\int_{-\infty}^{+\infty} P(A \mid x = x) p(x) dx}. \quad (28)$$

1.3 Expectation, Variance, and Random Vectors

1.3.1 Mean

The expected value (mean) of a continuous random variable, and its conditional version, are

$$m_x = \mathbb{E}(x) \quad (29)$$

$$= \int_{-\infty}^{+\infty} x p(x) dx, \quad (30)$$

$$m_{x|A} = \mathbb{E}(x \mid A) \quad (31)$$

$$= \int_{-\infty}^{+\infty} x p(x \mid A) dx. \quad (32)$$

1.3.2 Variance

$$\sigma_x^2 = \mathbb{E}((x - m_x)^2) \quad (33)$$

$$= \int_{-\infty}^{+\infty} (x - m_x)^2 p(x) dx \quad (34)$$

$$= \mathbb{E}(x^2) - (\mathbb{E}(x))^2, \quad (35)$$

with the conditional variance

$$\sigma_{x|A}^2 = \mathbb{E}\{(x - m_{x|A})^2 \mid A\} \quad (36)$$

$$= \int_{-\infty}^{+\infty} (x - m_{x|A})^2 p(x \mid A) dx. \quad (37)$$

1.3.3 A set of random variables

The joint pdf of $x_1, \dots, x_n$ satisfies

$$\int_{-\infty}^{+\infty} \cdots \int_{-\infty}^{+\infty} p(x_1, \dots, x_n) dx_1 \cdots dx_n = 1. \quad (38)$$

If x_i, x_j are independent, $p(x_i, x_j) = p(x_i)p(x_j)$. Expectation is linear: for $y = \alpha x_i + \beta x_j$,

$$\mathbb{E}(y) = \alpha \mathbb{E}(x_i) + \beta \mathbb{E}(x_j). \quad (39)$$

1.3.4 Covariance

$$\sigma_{ij}^2 = \mathbb{E}((x_i - m_i)(x_j - m_j)) \quad (40)$$

$$= \mathbb{E}(x_i x_j) - \mathbb{E}(x_i) \mathbb{E}(x_j). \quad (41)$$

When $\sigma_{ij} = 0$ we have $\mathbb{E}(x_i x_j) = \mathbb{E}(x_i) \mathbb{E}(x_j)$ and x_i, x_j are *uncorrelated*.

Discussion questions.

1. If x_i, x_j are independent, then $p(x_i, x_j) = p(x_i)p(x_j) \Rightarrow \mathbb{E}(x_i x_j) = \mathbb{E}(x_i)\mathbb{E}(x_j)$ — so independence implies uncorrelated.
2. If x_i, x_j are uncorrelated, are they necessarily independent?
3. If x_i, x_j are jointly Gaussian, is independence equivalent to being uncorrelated?

1.3.5 Random vector

Let $\mathbf{x} = [x_1, \dots, x_n]^\top$, where all elements are random variables, with mean $\mathbb{E}(\mathbf{x}) \triangleq \mathbf{m}_x$ and covariance

$$\mathbf{P} = \mathbb{E}((\mathbf{x} - \mathbf{m}_x)(\mathbf{x} - \mathbf{m}_x)^\top) \quad (42)$$

$$= \mathbb{E}(\mathbf{x}\mathbf{x}^\top) - \mathbb{E}(\mathbf{x}) \mathbb{E}(\mathbf{x}^\top), \quad (43)$$

with entries

$$P_{ii} = \mathbb{E}((x_i - m_i)^2) \quad (44)$$

$$= \int_{-\infty}^{+\infty} (x_i - m_i)^2 p(x_i) dx_i, \quad (45)$$

$$P_{ij} = \mathbb{E}((x_i - m_i)(x_j - m_j)). \quad (46)$$

Since $P_{ij} = P_{ji}$, the covariance $\mathbf{P}$ is a symmetric matrix; it is also positive (semi-)definite.

1.3.6 Gaussian vector pdf

$$p(\mathbf{x}) = \frac{1}{(2\pi)^{n/2} \sqrt{|\mathbf{P}|}} \exp\left(-\frac{1}{2}(\mathbf{x} - \mathbf{m}_x)^\top \mathbf{P}^{-1}(\mathbf{x} - \mathbf{m}_x)\right), \quad (47)$$

where $|\mathbf{P}|$ is the determinant of $\mathbf{P}$, $\mathbf{m}_x = \mathbb{E}(\mathbf{x})$, and $\mathbf{P} = \mathbb{E}((\mathbf{x} - \mathbf{m}_x)(\mathbf{x} - \mathbf{m}_x)^\top)$.

Homework. Let $\mathbf{x} = [x_1, \dots, x_n]^\top$ with $p(\mathbf{x}) = \mathcal{N}(\mathbf{m}_x, \mathbf{P}_{xx})$. Define

$$\begin{aligned} q &= (\mathbf{x} - \mathbf{m}_x)^\top \mathbf{P}_{xx}^{-1}(\mathbf{x} - \mathbf{m}_x), \\ \mathbf{y} &= \mathbf{P}_{xx}^{-1/2}(\mathbf{x} - \mathbf{m}_x), \\ q &= \mathbf{y}^\top \mathbf{y} = \sum_i y_i^2. \end{aligned}$$

What are the mean and covariance of q ?

1.4 Maximum Likelihood and Maximum a Posteriori Estimation

For a measurement model with additive noise $z_i = h_i(\mathbf{x}) + n_i$, stacked as $\mathbf{z} = \mathbf{h}(\mathbf{x}) + \mathbf{n}$, two estimators are

- **Maximum likelihood estimation (MLE):** $\max_{\mathbf{x}} p(\mathbf{z} | \mathbf{x})$.
- **Maximum a posteriori (MAP):** $\max_{\mathbf{x}} p(\mathbf{x} | \mathbf{z})$, with $p(\mathbf{x} | \mathbf{z}) = \frac{p(\mathbf{z} | \mathbf{x}) p(\mathbf{x})}{p(\mathbf{z})}$.

For zero-mean Gaussian noise $\mathbf{n} \sim \mathcal{N}(\mathbf{0}, \mathbf{P})$,

$$\max_{\mathbf{x}} p(\mathbf{z} | \mathbf{x}) = \max_{\mathbf{x}} \frac{1}{(2\pi)^{n/2} \sqrt{|\mathbf{P}|}} \exp\left(-(\mathbf{z} - \mathbf{h}(\mathbf{x}))^\top \mathbf{P}^{-1}(\mathbf{z} - \mathbf{h}(\mathbf{x}))\right), \quad (48)$$

which is equivalent to the weighted least-squares problem

$$\min_{\mathbf{x}} C = (\mathbf{z} - \mathbf{h}(\mathbf{x}))^\top \mathbf{P}^{-1}(\mathbf{z} - \mathbf{h}(\mathbf{x})). \quad (49)$$

Linearizing about $\hat{\mathbf{x}}$ with $\mathbf{z} = \mathbf{h}(\mathbf{x}) + \mathbf{n} = \mathbf{h}(\hat{\mathbf{x}}) + \mathbf{H}(\mathbf{x} - \hat{\mathbf{x}}) + \mathbf{n}$, and defining the residual $\tilde{\mathbf{z}} = \mathbf{z} - \mathbf{h}(\hat{\mathbf{x}})$ and error $\tilde{\mathbf{x}} = \mathbf{x} - \hat{\mathbf{x}}$,

$$\min_{\tilde{\mathbf{x}}} (\tilde{\mathbf{z}} - \mathbf{H}\tilde{\mathbf{x}})^\top \mathbf{P}^{-1}(\tilde{\mathbf{z}} - \mathbf{H}\tilde{\mathbf{x}}) = \min_{\tilde{\mathbf{x}}} (\tilde{\mathbf{z}}^\top \tilde{\mathbf{z}} - 2\tilde{\mathbf{z}}^\top \mathbf{P}^{-1}\mathbf{H}\tilde{\mathbf{x}} + \tilde{\mathbf{x}}^\top \mathbf{H}^\top \mathbf{P}^{-1}\mathbf{H}\tilde{\mathbf{x}}). \quad (50)$$

Setting the gradient to zero gives the normal equations $\mathbf{H}^\top \mathbf{P}^{-1}\mathbf{H}\tilde{\mathbf{x}} = \mathbf{H}^\top \mathbf{P}^{-1}\tilde{\mathbf{z}}$, hence

$$\boxed{\begin{aligned} \mathbf{x}^\oplus &= \hat{\mathbf{x}} + \tilde{\mathbf{x}} \\ &= \hat{\mathbf{x}} + (\mathbf{H}^\top \mathbf{P}^{-1}\mathbf{H})^{-1}\mathbf{H}^\top \mathbf{P}^{-1}\tilde{\mathbf{z}}. \end{aligned}} \quad \begin{aligned} (51) \\ (52) \end{aligned}$$

This is the classical weighted least-squares solution.

Questions.

1. What is the covariance of the new estimate $\mathbf{x}^\oplus$?
2. For a prior $\mathbf{x}_p = \mathbf{x} + \mathbf{n}_p$ together with $z_i = h_i(\mathbf{x}) + n_i$, $i = 1, \dots, n$, what is the MAP solution $\max_{\mathbf{x}} p(\mathbf{x} | \mathbf{z})$?

1.4.1 Optimization

Nonlinear least-squares problems are solved iteratively, e.g. Gauss–Newton (GN) and Levenberg–Marquardt (LM).

1.5 Linear Systems

A linear (continuous-time) state-space system is

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u}, \quad (53)$$

$$\mathbf{y} = \mathbf{C}\mathbf{x} + \mathbf{D}\mathbf{u}, \quad (54)$$

where $\mathbf{x}$ is the state, $\mathbf{u}$ the input, and $\mathbf{y}$ the output. Its discrete-time counterpart is

$$\mathbf{x}_{k+1} = \mathbf{A}'\mathbf{x}_k + \mathbf{B}'\mathbf{u}_k, \quad (55)$$

$$\mathbf{y}_{k+1} = \mathbf{C}'\mathbf{x}_{k+1} + \mathbf{D}'\mathbf{u}_{k+1}. \quad (56)$$

If $\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D}$ are constant, the system is *linear time-invariant* (LTI); if they depend on time, it is *linear time-variant* (LTV).

1.5.1 Homogeneous solution

Consider $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$ with $\mathbf{x}(t_0) = \mathbf{x}_0$ and $\mathbf{y} = \mathbf{C}\mathbf{x}$. From $\dot{\mathbf{x}} - \mathbf{A}\mathbf{x} = \mathbf{0}$,

$$\frac{d}{dt}(e^{-\mathbf{A}t}\mathbf{x}) = e^{-\mathbf{A}t}\dot{\mathbf{x}} + e^{-\mathbf{A}t}(-\mathbf{A})\mathbf{x} \quad (57)$$

$$= e^{-\mathbf{A}t}(\dot{\mathbf{x}} - \mathbf{A}\mathbf{x}) \quad (58)$$

$$= \mathbf{0}. \quad (59)$$

Integrating from 0 to t gives $e^{-\mathbf{A}t}\mathbf{x}(t) - \mathbf{x}(t_0) = \mathbf{0}$, so

$$\mathbf{x}(t) = e^{\mathbf{A}t}\mathbf{x}_0 \quad (60)$$

$$= \Phi(t, 0)\mathbf{x}_0, \quad (61)$$

$$\Phi(t, 0) = e^{\mathbf{A}t}, \quad (62)$$

where $\Phi(t, 0)$ is the *state transition matrix*, and $\mathbf{y} = \mathbf{C}\Phi(t, 0)\mathbf{x}_0$.

1.5.2 Properties of the state transition matrix

$$\Phi(t_2, t_0) = \Phi(t_2, t_1)\Phi(t_1, t_0), \quad (63)$$

$$\frac{d}{dt}\Phi(t, 0) = \mathbf{A}\Phi(t, 0). \quad (64)$$

The second follows from $\dot{\mathbf{x}} = \dot{\Phi}(t, 0)\mathbf{x}_0 = \mathbf{A}\mathbf{x} = \mathbf{A}\Phi(t, 0)\mathbf{x}_0$.

1.5.3 Output sequence and observability

Sampling the homogeneous response at $t_0, t_1, \dots, t_n$ with $\mathbf{x}_k = \Phi(k, 0)\mathbf{x}_0$ and $\mathbf{y}_k = \mathbf{C}\mathbf{x}_k$ stacks into

$$\begin{bmatrix} \mathbf{y}_1 \\ \vdots \\ \mathbf{y}_n \end{bmatrix} = \begin{bmatrix} \mathbf{C}\Phi(1, 0) \\ \vdots \\ \mathbf{C}\Phi(n, 0) \end{bmatrix} \mathbf{x}_0, \quad (65)$$

which relates the measured outputs to the initial state (an observability-type relation).

1.5.4 Full LTV solution

For the complete system $\dot{\mathbf{x}} = \mathbf{A}(t)\mathbf{x} + \mathbf{B}(t)\mathbf{u}(t)$, write $\dot{\mathbf{x}} - \mathbf{A}(t)\mathbf{x} = \mathbf{B}(t)\mathbf{u}(t)$ and use the integrating factor $e^{-\int_0^t \mathbf{A}(\tau) d\tau}$:

$$\frac{d}{dt} \left(e^{-\int_0^t \mathbf{A}(\tau) d\tau} \mathbf{x} \right) = e^{-\int_0^t \mathbf{A}(\tau) d\tau} (\dot{\mathbf{x}} - \mathbf{A}(t)\mathbf{x}) \quad (66)$$

$$= e^{-\int_0^t \mathbf{A}(\tau) d\tau} \mathbf{B}(t)\mathbf{u}(t). \quad (67)$$

Integrating,

$$e^{-\int_0^t \mathbf{A}(\tau) d\tau} \mathbf{x}(t) - \mathbf{x}_0 = \int_0^t e^{-\int_0^\tau \mathbf{A}(s) ds} \mathbf{B}(\tau)\mathbf{u}(\tau) d\tau, \quad (68)$$

so that, with $\Phi(t, 0) = e^{\int_0^t \mathbf{A}(\tau) d\tau}$ and $\Phi(t, \tau) = e^{\int_\tau^t \mathbf{A}(s) ds}$,

$$\mathbf{x}(t) = \Phi(t, 0) \mathbf{x}_0 + \int_0^t \Phi(t, \tau) \mathbf{B}(\tau) \mathbf{u}(\tau) d\tau, \quad (69)$$

$$\mathbf{y}(t) = \mathbf{C}(t) \left(\Phi(t, 0) \mathbf{x}_0 + \int_0^t \Phi(t, \tau) \mathbf{B}(\tau) \mathbf{u}(\tau) d\tau \right) + \mathbf{D}(t)\mathbf{u}(t). \quad (70)$$

This is one of the most important properties of linear systems.

1.6 Homework Problems and Solutions

1.6.1 Variance identity

Problem. Show that the variance of a scalar random variable can be written as $\sigma_x^2 = \mathbb{E}(\mathbf{x}^2) - (\mathbb{E}(\mathbf{x}))^2$, and discuss when it vanishes.

Solution. Expanding the definition with $m_x = \mathbb{E}(\mathbf{x})$,

$$\sigma_x^2 = \mathbb{E}((\mathbf{x} - m_x)^2) \quad (71)$$

$$= \mathbb{E}(\mathbf{x}^2 - 2m_x\mathbf{x} + m_x^2) \quad (72)$$

$$= \mathbb{E}(\mathbf{x}^2) - 2m_x\mathbb{E}(\mathbf{x}) + m_x^2 \quad (73)$$

$$= \mathbb{E}(\mathbf{x}^2) - m_x^2. \quad (74)$$

Since $\sigma_x^2 \geq 0$, we have $\mathbb{E}(\mathbf{x}^2) \geq (\mathbb{E}(\mathbf{x}))^2$, with equality if and only if $\sigma_x^2 = 0$, i.e. $\mathbf{x} = m_x$ almost surely (a deterministic variable) — a degenerate special case.

1.6.2 Covariance matrix identity

Problem. For a random vector $\mathbf{x}$ with mean $\mathbf{m}_x$, show that $\mathbf{P} = \mathbb{E}((\mathbf{x} - \mathbf{m}_x)(\mathbf{x} - \mathbf{m}_x)^\top) = \mathbb{E}(\mathbf{x}\mathbf{x}^\top) - \mathbb{E}(\mathbf{x})\mathbb{E}(\mathbf{x}^\top)$.

Solution. Expanding the outer product and using linearity of expectation, together with $\mathbb{E}(\mathbf{x}) = \mathbf{m}_x$,

$$\mathbf{P} = \mathbb{E}(\mathbf{x}\mathbf{x}^\top - \mathbf{x}\mathbf{m}_x^\top - \mathbf{m}_x\mathbf{x}^\top + \mathbf{m}_x\mathbf{m}_x^\top) \quad (75)$$

$$= \mathbb{E}(\mathbf{x}\mathbf{x}^\top) - \mathbb{E}(\mathbf{x})\mathbf{m}_x^\top - \mathbf{m}_x\mathbb{E}(\mathbf{x}^\top) + \mathbf{m}_x\mathbf{m}_x^\top \quad (76)$$

$$= \mathbb{E}(\mathbf{x}\mathbf{x}^\top) - \mathbf{m}_x\mathbf{m}_x^\top - \mathbf{m}_x\mathbf{m}_x^\top + \mathbf{m}_x\mathbf{m}_x^\top \quad (77)$$

$$= \mathbb{E}(\mathbf{x}\mathbf{x}^\top) - \mathbf{m}_x\mathbf{m}_x^\top \quad (78)$$

$$= \mathbb{E}(\mathbf{x}\mathbf{x}^\top) - \mathbb{E}(\mathbf{x})\mathbb{E}(\mathbf{x}^\top). \quad (79)$$

1.6.3 Scalar covariance identity

Problem. For two random variables x_i, x_j , show that $\sigma_{ij} = \mathbb{E}(x_i x_j) - \mathbb{E}(x_i)\mathbb{E}(x_j)$.

Solution. With $m_i = \mathbb{E}(x_i)$, $m_j = \mathbb{E}(x_j)$, expand

$$\sigma_{ij} = \mathbb{E}((x_i - m_i)(x_j - m_j)) \quad (80)$$

$$= \iint (x_i - m_i)(x_j - m_j) p(x_i, x_j) dx_i dx_j \quad (81)$$

$$\begin{aligned} &= \iint x_i x_j p(x_i, x_j) dx_i dx_j - m_j \iint x_i p(x_i, x_j) dx_i dx_j \\ &\quad - m_i \iint x_j p(x_i, x_j) dx_i dx_j + m_i m_j \iint p(x_i, x_j) dx_i dx_j. \end{aligned} \quad (82)$$

Using the marginalization identities (and symmetrically for x_j)

$$\int p(x_i, x_j) dx_i = p(x_j), \quad \iint p(x_i, x_j) dx_i dx_j = 1, \quad (83)$$

we get

$$m_i \iint x_j p(x_i, x_j) dx_i dx_j = m_i \int x_j p(x_j) dx_j \quad (84)$$

$$= m_i m_j, \quad (85)$$

and likewise for the other term, so

$$\sigma_{ij} = \mathbb{E}(x_i x_j) - m_i m_j - m_i m_j + m_i m_j \quad (86)$$

$$= \mathbb{E}(x_i x_j) - \mathbb{E}(x_i)\mathbb{E}(x_j). \quad (87)$$

1.6.4 Independence versus uncorrelatedness

Problem. Resolve the three discussion questions: (1) does independence imply uncorrelatedness? (2) does uncorrelatedness imply independence? (3) for jointly Gaussian variables, are the two equivalent?

Solution. (1) *Independent $\Rightarrow$ uncorrelated.* If $p(x_i, x_j) = p(x_i)p(x_j)$, then

$$\mathbb{E}(x_i x_j) = \iint x_i x_j p(x_i, x_j) dx_i dx_j \quad (88)$$

$$= \int x_i p(x_i) dx_i \cdot \int x_j p(x_j) dx_j \quad (89)$$

$$= \mathbb{E}(x_i)\mathbb{E}(x_j), \quad (90)$$

so $\sigma_{ij} = 0$.

(2) *Uncorrelated $\nRightarrow$ independent.* Uncorrelatedness gives only

$$\mathbb{E}(x_i x_j) - \mathbb{E}(x_i)\mathbb{E}(x_j) = \iint x_i x_j (p(x_i, x_j) - p(x_i)p(x_j)) dx_i dx_j \quad (91)$$

$$= 0, \quad (92)$$

which does *not* force $p(x_i, x_j) = p(x_i)p(x_j)$ in general. Hence uncorrelated does not imply independence.

(3) *Jointly Gaussian.* For $x_i \sim \mathcal{N}(m_i, \sigma_i^2)$ and $x_j \sim \mathcal{N}(m_j, \sigma_j^2)$, the product of marginals is

$$p(x_i)p(x_j) = \frac{1}{2\pi\sigma_i\sigma_j} \exp\left\{-\frac{1}{2} \begin{bmatrix} x_i - m_i \\ x_j - m_j \end{bmatrix}^\top \begin{bmatrix} \sigma_i^2 & 0 \\ 0 & \sigma_j^2 \end{bmatrix}^{-1} \begin{bmatrix} x_i - m_i \\ x_j - m_j \end{bmatrix}\right\}. \quad (93)$$

The joint Gaussian $[x_i, x_j]^\top \sim \mathcal{N}([m_i, m_j]^\top, \mathbf{P}_{xx})$ with $\mathbf{P}_{xx} = \begin{bmatrix} \sigma_i^2 & \sigma_{ij} \\ \sigma_{ji} & \sigma_j^2 \end{bmatrix}$ is

$$p(x_i, x_j) = \frac{1}{2\pi\sqrt{|\mathbf{P}_{xx}|}} \exp\left\{-\frac{1}{2} \begin{bmatrix} x_i - m_i \\ x_j - m_j \end{bmatrix}^\top \mathbf{P}_{xx}^{-1} \begin{bmatrix} x_i - m_i \\ x_j - m_j \end{bmatrix}\right\}. \quad (94)$$

If $\sigma_{ij} = \sigma_{ji} = 0$, then $\mathbf{P}_{xx}$ is diagonal and $p(x_i, x_j) = p(x_i)p(x_j)$. Hence, for (jointly) Gaussian variables, *independent* $\Leftrightarrow$ *uncorrelated*.

1.6.5 The χ^2 distribution

Problem. Let $\mathbf{x} = [x_1, \dots, x_n]^\top$ with $p(\mathbf{x}) = \mathcal{N}(\mathbf{m}_x, \mathbf{P}_{xx})$ and define $q = (\mathbf{x} - \mathbf{m}_x)^\top \mathbf{P}_{xx}^{-1} (\mathbf{x} - \mathbf{m}_x)$. Find the mean and variance of q .

Solution. Factor $\mathbf{P}_{xx} = \mathbf{L}\mathbf{L}^\top$ (Cholesky), so $\mathbf{P}_{xx}^{-1} = \mathbf{L}^{-\top}\mathbf{L}^{-1}$, and define the whitened variable $\mathbf{y} = \mathbf{L}^{-1}(\mathbf{x} - \mathbf{m}_x)$. Then

$$q = (\mathbf{x} - \mathbf{m}_x)^\top \mathbf{L}^{-\top} \mathbf{L}^{-1} (\mathbf{x} - \mathbf{m}_x) \quad (95)$$

$$= \mathbf{y}^\top \mathbf{y} \quad (96)$$

$$= \sum_{i=1}^n y_i^2. \quad (97)$$

The whitened variable has zero mean and identity covariance,

$$\mathbf{m}_y = \mathbb{E}(\mathbf{y}) \quad (98)$$

$$= \mathbf{L}^{-1}(\mathbb{E}(\mathbf{x}) - \mathbf{m}_x) \quad (99)$$

$$= \mathbf{0}, \quad (100)$$

$$\mathbf{P}_{yy} = \mathbb{E}(\mathbf{y}\mathbf{y}^\top) \quad (101)$$

$$= \mathbf{L}^{-1} \mathbb{E}((\mathbf{x} - \mathbf{m}_x)(\mathbf{x} - \mathbf{m}_x)^\top) \mathbf{L}^{-\top} \quad (102)$$

$$= \mathbf{L}^{-1} \mathbf{P}_{xx} \mathbf{L}^{-\top} \quad (103)$$

$$= \mathbf{L}^{-1} \mathbf{L}\mathbf{L}^\top \mathbf{L}^{-\top} \quad (104)$$

$$= \mathbf{I}_n, \quad (105)$$

so $y_i \sim \mathcal{N}(0, 1)$ are independent. Therefore the mean of q is

$$\mathbb{E}(q) = \mathbb{E}(\mathbf{y}^\top \mathbf{y}) \quad (106)$$

$$= \sum_{i=1}^n \mathbb{E}(y_i^2) \quad (107)$$

$$= n. \quad (108)$$

For the variance, with $\mathbb{E}(y_i^2) = 1$ the cross terms vanish by independence, giving

$$\sigma_{qq}^2 = \mathbb{E}((q - n)^2) \quad (109)$$

$$= \mathbb{E}\left(\left(\sum_i (y_i^2 - 1)\right)^2\right) \quad (110)$$

$$= \sum_i \mathbb{E}((y_i^2 - 1)^2) \quad (111)$$

$$= \sum_i (\mathbb{E}(y_i^4) - 2\mathbb{E}(y_i^2) + 1). \quad (112)$$

The fourth moment of a standard normal follows from integration by parts (with $u = y_i^3$, $dv = y_i p(y_i) dy_i$, $v = -p(y_i)$):

$$\mathbb{E}(y_i^4) = \int_{-\infty}^{+\infty} y_i^4 p(y_i) dy_i \quad (113)$$

$$= \underbrace{[-y_i^3 p(y_i)]_{-\infty}^{+\infty}}_{=0} + \int_{-\infty}^{+\infty} 3y_i^2 p(y_i) dy_i \quad (114)$$

$$= 3 \mathbb{E}(y_i^2) \quad (115)$$

$$= 3. \quad (116)$$

Hence each term equals $3 - 2 + 1 = 2$, and

$$\sigma_{qq}^2 = \sum_{i=1}^n 2 = 2n. \quad (117)$$

So $q \sim \chi^2(n)$ has mean n and variance $2n$.

2 Lecture 2: Rotation, Lie Groups, SE(3), SE₂(3), and Quaternions

Further reading. This lecture draws on several standard references: matrix Lie groups in Barfoot [1, Ch. 7]; the Lie-group treatment of $SO(3)/SE(3)$ in the GTSAM notes [2], Atanasov's ECE276A slides [3], and the micro Lie theory of Solà, Deray & Atchuthan [4]; and quaternion algebra/kinematics in Trawny & Roumeliotis [5, Sec. 1] and Solà [6, Sec. 1–3].

2.1 Preliminaries: useful series

The matrix exponential and the sine/cosine series underpin everything that follows:

$$e^x = \exp(x) = \sum_{k=0}^{\infty} \frac{1}{k!} x^k \quad (118)$$

$$= I + x + \frac{1}{2!}x^2 + \frac{1}{3!}x^3 + \frac{1}{4!}x^4 + \dots, \quad (119)$$

$$\sin(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)!} x^{2k+1} \quad (120)$$

$$= x - \frac{1}{3!}x^3 + \frac{1}{5!}x^5 - \dots, \quad (121)$$

$$\cos(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k)!} x^{2k} \quad (122)$$

$$= 1 - \frac{1}{2!}x^2 + \frac{1}{4!}x^4 - \dots. \quad (123)$$

For small x (so that $x^2, x^3 \rightarrow 0$), the important first-order approximations are $e^x \approx I + x$, $\sin(x) \approx x$, and $\cos(x) \approx 1$.

2.2 Frames, notation, and transformations

We work with coordinate frames $\{A\}$ and $\{B\}$. A point/vector $\mathbf{L}$ expressed in $\{A\}$ is ${}^A\mathbf{L}$, and in $\{B\}$ is ${}^B\mathbf{L}$. The angle $\theta_{A \rightarrow B}$ is the rotation about the z -axis taking $\{A\}$ to $\{B\}$ (Fig. 1).

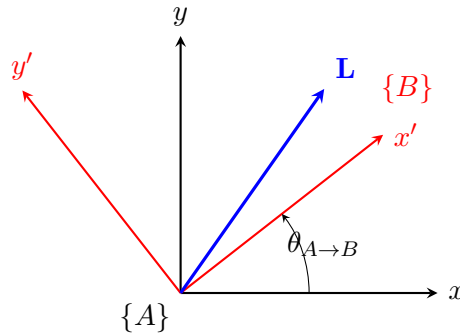


Figure 1: Frame $\{B\}$ is frame $\{A\}$ rotated by $\theta_{A \rightarrow B}$ about the z -axis; the vector $\mathbf{L}$ can be expressed in either frame.

The rotation matrix ${}^A_B\mathbf{R}$ maps a vector from $\{B\}$ to $\{A\}$:

$${}^A\mathbf{L} = {}^A_B\mathbf{R} {}^B\mathbf{L} \quad (124)$$

$$= {}^A_B\mathbf{R}(\theta_{A \rightarrow B}) {}^B\mathbf{L}. \quad (125)$$

Adding the translation ${}^A\mathbf{p}_B$ (the origin of $\{B\}$ expressed in $\{A\}$, Fig. 2) gives the full rigid-body transformation

$$\{A\}, \{B\} \Rightarrow \{{}_B^A\mathbf{R}, {}^A\mathbf{p}_B\} \Rightarrow {}_B^A\mathbf{T} = \begin{bmatrix} {}_B^A\mathbf{R} & {}^A\mathbf{p}_B \\ \mathbf{0}_{1 \times 3} & 1 \end{bmatrix}. \quad (126)$$

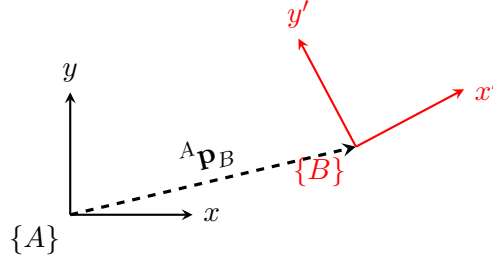


Figure 2: A rigid-body transformation combines the rotation ${}_B^A\mathbf{R}$ with the translation ${}^A\mathbf{p}_B$.

2.3 Defining rotation

2.3.1 Euler angles

With roll θ (about x), pitch r (about y), yaw ψ (about z),

$$\mathbf{R}(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{bmatrix}, \quad \mathbf{R}(r) = \begin{bmatrix} \cos r & 0 & \sin r \\ 0 & 1 & 0 \\ -\sin r & 0 & \cos r \end{bmatrix}, \quad \mathbf{R}(\psi) = \begin{bmatrix} \cos \psi & -\sin \psi & 0 \\ \sin \psi & \cos \psi & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad (127)$$

and the yaw–pitch–roll composition is

$$\mathbf{R} = \mathbf{R}(\psi) \mathbf{R}(r) \mathbf{R}(\theta) \quad (128)$$

$$= \begin{bmatrix} \cos \psi & -\sin \psi & 0 \\ \sin \psi & \cos \psi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos r & 0 & \sin r \\ 0 & 1 & 0 \\ -\sin r & 0 & \cos r \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{bmatrix}. \quad (129)$$

2.3.2 Infinitesimal rotation and the skew-symmetric matrix

For small angles $\delta\theta, \delta r, \delta\psi \rightarrow 0$ each elementary rotation linearizes ($\cos \rightarrow 1$, $\sin \rightarrow \text{angle}$):

$$\mathbf{R}(\delta\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -\delta\theta \\ 0 & \delta\theta & 1 \end{bmatrix}, \quad \mathbf{R}(\delta r) = \begin{bmatrix} 1 & 0 & \delta r \\ 0 & 1 & 0 \\ -\delta r & 0 & 1 \end{bmatrix}, \quad \mathbf{R}(\delta\psi) = \begin{bmatrix} 1 & -\delta\psi & 0 \\ \delta\psi & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}. \quad (130)$$

Multiplying the last two and dropping all second-order products ($\delta r \delta\theta \approx 0$ etc.),

$$\mathbf{R}(\delta r) \mathbf{R}(\delta\theta) = \begin{bmatrix} 1 & 0 & \delta r \\ 0 & 1 & 0 \\ -\delta r & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -\delta\theta \\ 0 & \delta\theta & 1 \end{bmatrix} \quad (131)$$

$$= \begin{bmatrix} 1 & 0 & \delta r \\ 0 & 1 & -\delta\theta \\ -\delta r & \delta\theta & 1 \end{bmatrix}, \quad (132)$$

and then

$$\begin{aligned}\mathbf{R} &= \mathbf{R}(\delta\psi)\mathbf{R}(\delta r)\mathbf{R}(\delta\theta) = \begin{bmatrix} 1 & -\delta\psi & 0 \\ \delta\psi & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & \delta r \\ 0 & 1 & -\delta\theta \\ -\delta r & \delta\theta & 1 \end{bmatrix} \\ &= \begin{bmatrix} 1 & -\delta\psi & \delta r \\ \delta\psi & 1 & -\delta\theta \\ -\delta r & \delta\theta & 1 \end{bmatrix} \end{aligned} \quad (133)$$

$$= \mathbf{I} + \begin{bmatrix} 0 & -\delta\psi & \delta r \\ \delta\psi & 0 & -\delta\theta \\ -\delta r & \delta\theta & 0 \end{bmatrix} \quad (134)$$

$$= \mathbf{I} + [\boldsymbol{\theta}]_{\times}, \quad (135)$$

where the small-angle vector is $\boldsymbol{\theta} = [\delta\theta, \delta r, \delta\psi]^\top$. The matrix $[\boldsymbol{\theta}]_{\times}$ is skew-symmetric, $[\boldsymbol{\theta}]_{\times} = -[\boldsymbol{\theta}]_{\times}^\top$. This motivates writing a finite rotation as $\mathbf{R}(\boldsymbol{\theta}) \approx \mathbf{I} + [\boldsymbol{\theta}]_{\times} \approx \exp([\boldsymbol{\theta}]_{\times})$, and defining $\text{Exp}(\boldsymbol{\theta}) \triangleq \exp([\boldsymbol{\theta}]_{\times})$. The question is then whether $\mathbf{R}(\boldsymbol{\theta}) = \text{Exp}(\boldsymbol{\theta})$.

2.4 Axis-angle and Rodrigues' formula

Let $\mathbf{k}$ be the unit rotation axis and θ the rotation angle, with $\boldsymbol{\theta} = \theta\mathbf{k} = [\theta_1, \theta_2, \theta_3]^\top$, so

$$\mathbf{k} = \begin{bmatrix} k_1 \\ k_2 \\ k_3 \end{bmatrix}, \quad [\mathbf{k}]_{\times} = \begin{bmatrix} 0 & -k_3 & k_2 \\ k_3 & 0 & -k_1 \\ -k_2 & k_1 & 0 \end{bmatrix}, \quad [\boldsymbol{\theta}]_{\times} = \begin{bmatrix} 0 & -\theta_3 & \theta_2 \\ \theta_3 & 0 & -\theta_1 \\ -\theta_2 & \theta_1 & 0 \end{bmatrix}. \quad (136)$$

The skew operator on a unit vector satisfies the cyclic power identities

$$[\mathbf{k}]_{\times}^2 = \mathbf{k}\mathbf{k}^\top - \mathbf{I}_3, \quad [\mathbf{k}]_{\times}^3 = -[\mathbf{k}]_{\times}, \quad [\mathbf{k}]_{\times}^4 = -[\mathbf{k}]_{\times}^2, \quad (137)$$

$$[\mathbf{k}]_{\times}^5 = [\mathbf{k}]_{\times}, \quad [\mathbf{k}]_{\times}^6 = [\mathbf{k}]_{\times}^2, \quad [\mathbf{k}]_{\times}^7 = -[\mathbf{k}]_{\times}. \quad (138)$$

Substituting $[\boldsymbol{\theta}]_{\times} = \theta [\mathbf{k}]_{\times}$ into the exponential series term by term,

$$\begin{aligned}\text{Exp}(\boldsymbol{\theta}) &= \exp(\theta [\mathbf{k}]_{\times}) = \mathbf{I} + \theta [\mathbf{k}]_{\times} + \frac{\theta^2}{2!} [\mathbf{k}]_{\times}^2 + \frac{\theta^3}{3!} [\mathbf{k}]_{\times}^3 + \frac{\theta^4}{4!} [\mathbf{k}]_{\times}^4 \\ &\quad + \frac{\theta^5}{5!} [\mathbf{k}]_{\times}^5 + \frac{\theta^6}{6!} [\mathbf{k}]_{\times}^6 + \frac{\theta^7}{7!} [\mathbf{k}]_{\times}^7 + \cdots \\ &= \mathbf{I} + \theta [\mathbf{k}]_{\times} + \frac{\theta^2}{2!} [\mathbf{k}]_{\times}^2 - \frac{\theta^3}{3!} [\mathbf{k}]_{\times} - \frac{\theta^4}{4!} [\mathbf{k}]_{\times}^2 \\ &\quad + \frac{\theta^5}{5!} [\mathbf{k}]_{\times} + \frac{\theta^6}{6!} [\mathbf{k}]_{\times}^2 - \frac{\theta^7}{7!} [\mathbf{k}]_{\times} + \cdots \\ &= \mathbf{I} + \underbrace{\left(\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \frac{\theta^7}{7!} + \cdots \right)}_{\sin \theta} [\mathbf{k}]_{\times} \\ &\quad + \underbrace{\left(\frac{\theta^2}{2!} - \frac{\theta^4}{4!} + \frac{\theta^6}{6!} - \cdots \right)}_{1 - \cos \theta} [\mathbf{k}]_{\times}^2, \end{aligned} \quad (139)$$

which gives **Rodrigues' rotation formula**

$$\boxed{\mathbf{R}(\boldsymbol{\theta}) = \text{Exp}(\boldsymbol{\theta}) = \exp(\theta [\mathbf{k}]_{\times}) = \mathbf{I} + \sin \theta [\mathbf{k}]_{\times} + (1 - \cos \theta) [\mathbf{k}]_{\times}^2.} \quad (140)$$

For a small angle $\delta\theta \rightarrow 0$, $\mathbf{R}(\delta\boldsymbol{\theta}) \approx \mathbf{I} + [\delta\boldsymbol{\theta}]_{\times}$ (small-angle approximation).

2.4.1 Logarithmic map

The log map is the inverse of the exponential map:

$$\log(\mathbf{R}) = [\boldsymbol{\theta}]_{\times}, \quad \text{Log}(\mathbf{R}) = \boldsymbol{\theta}_{3 \times 1}, \quad \theta = \arccos\left(\frac{\text{tr} \mathbf{R} - 1}{2}\right), \quad [\mathbf{k}]_{\times} = \frac{\mathbf{R} - \mathbf{R}^{\top}}{2 \sin \theta}. \quad (141)$$

2.5 SO(3) kinematics

The two directions of the maps are: from the vector space $\mathbb{R}^3$ via the Lie algebra to the Lie group, $\boldsymbol{\theta} \rightarrow [\boldsymbol{\theta}]_{\times} \rightarrow \mathbf{R} = \text{Exp}(\boldsymbol{\theta})$; and back, $\mathbf{R} \rightarrow [\boldsymbol{\theta}]_{\times} \rightarrow \boldsymbol{\theta} = \text{Log}(\mathbf{R})$.

(1) Angular-velocity kinematics. Differentiating the orthogonality constraint $\mathbf{R}\mathbf{R}^{\top} = \mathbf{I}_3$,

$$\frac{d}{dt}(\mathbf{R}\mathbf{R}^{\top}) = \mathbf{0} \Rightarrow \dot{\mathbf{R}}\mathbf{R}^{\top} + \mathbf{R}\dot{\mathbf{R}}^{\top} = \mathbf{0} \Rightarrow \dot{\mathbf{R}}\mathbf{R}^{\top} = -\mathbf{R}\dot{\mathbf{R}}^{\top} = -(\dot{\mathbf{R}}\mathbf{R}^{\top})^{\top}, \quad (142)$$

so $\dot{\mathbf{R}}\mathbf{R}^{\top}$ is skew-symmetric. Writing it as $[\boldsymbol{\omega}]_{\times}$ with $[\boldsymbol{\omega}]_{\times} = \begin{bmatrix} 0 & -\omega_3 & \omega_2 \\ \omega_3 & 0 & -\omega_1 \\ -\omega_2 & \omega_1 & 0 \end{bmatrix}$,

$$\dot{\mathbf{R}}\mathbf{R}^{\top} = [\boldsymbol{\omega}]_{\times} \Rightarrow \dot{\mathbf{R}} = [\boldsymbol{\omega}]_{\times} \mathbf{R}. \quad (143)$$

(2) Equivalent limit form. With $\mathbf{R}(t + \Delta t) = \mathbf{R}(t) \exp([\boldsymbol{\omega}]_{\times} \Delta t)$,

$$\dot{\mathbf{R}}(t) = \lim_{\Delta t \rightarrow 0} \frac{\mathbf{R}(t + \Delta t) - \mathbf{R}(t)}{\Delta t} \quad (144)$$

$$= \lim_{\Delta t \rightarrow 0} \frac{\mathbf{R}(t) \exp([\boldsymbol{\omega}]_{\times} \Delta t) - \mathbf{R}(t)}{\Delta t} \quad (145)$$

$$= \mathbf{R}(t) \lim_{\Delta t \rightarrow 0} \frac{\exp([\boldsymbol{\omega}]_{\times} \Delta t) - \mathbf{I}}{\Delta t} \quad (146)$$

$$= \mathbf{R}(t) [\boldsymbol{\omega}]_{\times}. \quad (147)$$

(3) Integration. With $\mathbf{R}_i = \exp([\boldsymbol{\omega}_i]_{\times} \Delta t_i)$, composing increments gives

$$\mathbf{R}_{1 \rightarrow k} = \mathbf{R}_k \mathbf{R}_{k-1} \cdots \mathbf{R}_2 \mathbf{R}_1 \quad (148)$$

$$= \exp([\boldsymbol{\omega}_k]_{\times} \Delta t_k) \cdots \exp([\boldsymbol{\omega}_1]_{\times} \Delta t_1). \quad (149)$$

(4) Interpolation. For ${}^G_{t_1}\mathbf{R}, {}^G_{t_2}\mathbf{R}$ and $t \in [t_1, t_2]$, define the relative rotation and its angle, and the interpolation ratio:

$${}^{t_1}_{t_2}\mathbf{R} = {}^G_{t_1}\mathbf{R}^{\top} {}^G_{t_2}\mathbf{R}, \quad (150)$$

$${}^{t_1}_{t_2}\boldsymbol{\theta} = \text{Log}({}^{t_1}_{t_2}\mathbf{R}) = \text{Log}({}^G_{t_1}\mathbf{R}^{\top} {}^G_{t_2}\mathbf{R}), \quad (151)$$

$$\lambda = \frac{t - t_1}{t_2 - t_1}. \quad (152)$$

Then

$${}^G_t\mathbf{R} = {}^G_{t_1}\mathbf{R} {}^{t_1}_t\mathbf{R} \quad (153)$$

$$= {}^G_{t_1}\mathbf{R} \text{Exp}(\lambda {}^{t_1}_{t_2}\boldsymbol{\theta}). \quad (154)$$

(5) Right Jacobian. For a rotation $\mathbf{R}(\boldsymbol{\theta}) = \text{Exp}(\boldsymbol{\theta})$, perturb the angle by $\delta\boldsymbol{\theta} \in \mathbb{R}^3$ and let $\delta\boldsymbol{\varphi}$ be the corresponding small rotation in $\mathfrak{so}(3)$:

$$\mathbf{R}(\boldsymbol{\theta}) \text{Exp}(\delta\boldsymbol{\varphi}) = \mathbf{R}(\boldsymbol{\theta} + \delta\boldsymbol{\theta}) \Rightarrow \text{Exp}(\boldsymbol{\theta}) \text{Exp}(\delta\boldsymbol{\varphi}) = \text{Exp}(\boldsymbol{\theta} + \delta\boldsymbol{\theta}). \quad (155)$$

Solving for $\delta\boldsymbol{\varphi}$,

$$\text{Exp}(\delta\boldsymbol{\varphi}) = \text{Exp}(\boldsymbol{\theta})^{-1} \text{Exp}(\boldsymbol{\theta} + \delta\boldsymbol{\theta}) \Rightarrow \delta\boldsymbol{\varphi} = \text{Log}(\text{Exp}(\boldsymbol{\theta})^{-1} \text{Exp}(\boldsymbol{\theta} + \delta\boldsymbol{\theta})), \quad (156)$$

and the *right Jacobian* is the derivative of $\delta\boldsymbol{\varphi}$ with respect to $\delta\boldsymbol{\theta}$:

$$\mathbf{J}_r(\boldsymbol{\theta}) \triangleq \frac{\partial \delta\boldsymbol{\varphi}}{\partial \delta\boldsymbol{\theta}} = \lim_{\delta\boldsymbol{\theta} \rightarrow 0} \frac{\text{Log}(\text{Exp}(\boldsymbol{\theta})^{-1} \text{Exp}(\boldsymbol{\theta} + \delta\boldsymbol{\theta}))}{\delta\boldsymbol{\theta}}. \quad (157)$$

Equivalently, $\text{Exp}(\boldsymbol{\theta} + \delta\boldsymbol{\theta}) = \text{Exp}(\boldsymbol{\theta}) \text{Exp}(\mathbf{J}_r(\boldsymbol{\theta}) \delta\boldsymbol{\theta})$, and inverting,

$$\text{Exp}(\boldsymbol{\theta}) \text{Exp}(\delta\boldsymbol{\varphi}) = \text{Exp}(\boldsymbol{\theta} + \mathbf{J}_r^{-1}(\boldsymbol{\theta}) \delta\boldsymbol{\varphi}) \Rightarrow \text{Log}(\text{Exp}(\boldsymbol{\theta}) \text{Exp}(\delta\boldsymbol{\varphi})) = \boldsymbol{\theta} + \mathbf{J}_r^{-1}(\boldsymbol{\theta}) \delta\boldsymbol{\varphi}, \quad (158)$$

with the closed form

$$\mathbf{J}_r(\boldsymbol{\theta}) = \mathbf{I} - \frac{1 - \cos\|\boldsymbol{\theta}\|}{\|\boldsymbol{\theta}\|^2} [\boldsymbol{\theta}]_{\times} + \frac{\|\boldsymbol{\theta}\| - \sin\|\boldsymbol{\theta}\|}{\|\boldsymbol{\theta}\|^3} [\boldsymbol{\theta}]_{\times}^2. \quad (159)$$

(6) Left Jacobian. Applying the perturbation on the left instead defines the left Jacobian:

$$\text{Exp}(\boldsymbol{\theta} + \delta\boldsymbol{\theta}) = \text{Exp}(\boldsymbol{\theta}) \text{Exp}(\mathbf{J}_r(\boldsymbol{\theta}) \delta\boldsymbol{\theta}) = \text{Exp}(\mathbf{J}_l(\boldsymbol{\theta}) \delta\boldsymbol{\theta}) \text{Exp}(\boldsymbol{\theta}). \quad (160)$$

Using the adjoint identity $\text{Exp}(\boldsymbol{\theta}) \text{Exp}(\mathbf{x}) \text{Exp}(\boldsymbol{\theta})^{-1} = \text{Exp}(\mathbf{R}(\boldsymbol{\theta})\mathbf{x})$ on the middle expression,

$$\text{Exp}(\boldsymbol{\theta}) \text{Exp}(\mathbf{J}_r \delta\boldsymbol{\theta}) = \text{Exp}(\mathbf{R}(\boldsymbol{\theta})\mathbf{J}_r \delta\boldsymbol{\theta}) \text{Exp}(\boldsymbol{\theta}) \Rightarrow \mathbf{J}_l(\boldsymbol{\theta}) = \mathbf{R}(\boldsymbol{\theta}) \mathbf{J}_r(\boldsymbol{\theta}). \quad (161)$$

(7) Sign symmetry. From $\text{Exp}(\boldsymbol{\theta} + \delta\boldsymbol{\theta}) = \text{Exp}(\boldsymbol{\theta}) \text{Exp}(\mathbf{J}_r(\boldsymbol{\theta})\delta\boldsymbol{\theta})$, replacing $\boldsymbol{\theta} \rightarrow -\boldsymbol{\theta}$, $\delta\boldsymbol{\theta} \rightarrow -\delta\boldsymbol{\theta}$,

$$\text{Exp}(-\boldsymbol{\theta} - \delta\boldsymbol{\theta}) = \text{Exp}(-\mathbf{J}_l(-\boldsymbol{\theta})\delta\boldsymbol{\theta}) \text{Exp}(-\boldsymbol{\theta}) = \text{Exp}(-\mathbf{J}_r(\boldsymbol{\theta})\delta\boldsymbol{\theta}) \text{Exp}(-\boldsymbol{\theta}) \Rightarrow \mathbf{J}_l(-\boldsymbol{\theta}) = \mathbf{J}_r(\boldsymbol{\theta}). \quad (162)$$

(8) Summary. Collecting (5)–(7),

$$\text{Exp}(\boldsymbol{\theta} + \delta\boldsymbol{\theta}) = \text{Exp}(\boldsymbol{\theta}) \text{Exp}(\mathbf{J}_r(\boldsymbol{\theta})\delta\boldsymbol{\theta}), \quad \mathbf{J}_l(\boldsymbol{\theta}) = \mathbf{R}(\boldsymbol{\theta})\mathbf{J}_r(\boldsymbol{\theta}), \quad \mathbf{J}_l(-\boldsymbol{\theta}) = \mathbf{J}_r(\boldsymbol{\theta}). \quad (163)$$

(9) Identity $[\mathbf{R}\boldsymbol{\omega}]_{\times} = \mathbf{R} [\boldsymbol{\omega}]_{\times} \mathbf{R}^{\top}$. For any vector $\mathbf{v}$, using $\mathbf{R}(\mathbf{u} \times \mathbf{v}) = (\mathbf{R}\mathbf{u}) \times (\mathbf{R}\mathbf{v})$ and $\mathbf{R}^{\top}\mathbf{R} = \mathbf{I}$,

$$[\mathbf{R}\mathbf{u}]_{\times} (\mathbf{R}\mathbf{v}) = \mathbf{R} ([\mathbf{u}]_{\times} \mathbf{v}) \quad (164)$$

$$= \mathbf{R} ([\mathbf{u}]_{\times} \mathbf{R}^{\top} \mathbf{R}\mathbf{v}) \quad (165)$$

$$= \mathbf{R} [\mathbf{u}]_{\times} \mathbf{R}^{\top} (\mathbf{R}\mathbf{v}). \quad (166)$$

Since this holds for all $\mathbf{R}\mathbf{v}$, we conclude $[\mathbf{R}\mathbf{u}]_{\times} = \mathbf{R} [\mathbf{u}]_{\times} \mathbf{R}^{\top}$.

Homework. Prove $[\mathbf{R}\boldsymbol{\omega}]_{\times} = \mathbf{R} [\boldsymbol{\omega}]_{\times} \mathbf{R}^{\top}$.

(10) Identity $\text{Exp}(\mathbf{R}\boldsymbol{\varphi}) = \mathbf{R} \text{Exp}(\boldsymbol{\varphi}) \mathbf{R}^\top$. Using $[\mathbf{R}\boldsymbol{\varphi}]_\times^k = \mathbf{R} [\boldsymbol{\varphi}]_\times^k \mathbf{R}^\top$ (which follows from (9), since $[\mathbf{R}\boldsymbol{\varphi}]_\times^2 = \mathbf{R} [\boldsymbol{\varphi}]_\times \mathbf{R}^\top \mathbf{R} [\boldsymbol{\varphi}]_\times \mathbf{R}^\top = \mathbf{R} [\boldsymbol{\varphi}]_\times^2 \mathbf{R}^\top$, etc.),

$$\begin{aligned} \text{Exp}(\mathbf{R}\boldsymbol{\varphi}) &= \mathbf{I} + [\mathbf{R}\boldsymbol{\varphi}]_\times + \frac{1}{2!} [\mathbf{R}\boldsymbol{\varphi}]_\times^2 + \frac{1}{3!} [\mathbf{R}\boldsymbol{\varphi}]_\times^3 + \cdots \\ &= \mathbf{R}\mathbf{R}^\top + \mathbf{R} [\boldsymbol{\varphi}]_\times \mathbf{R}^\top + \frac{1}{2!} \mathbf{R} [\boldsymbol{\varphi}]_\times^2 \mathbf{R}^\top + \cdots \\ &= \mathbf{R}(\mathbf{I} + [\boldsymbol{\varphi}]_\times + \frac{1}{2!} [\boldsymbol{\varphi}]_\times^2 + \frac{1}{3!} [\boldsymbol{\varphi}]_\times^3 + \cdots) \mathbf{R}^\top \end{aligned} \quad (167)$$

$$= \mathbf{R} \text{Exp}(\boldsymbol{\varphi}) \mathbf{R}^\top. \quad (168)$$

Homework. Prove $\text{Exp}(\mathbf{R}\boldsymbol{\varphi}) = \mathbf{R} \text{Exp}(\boldsymbol{\varphi}) \mathbf{R}^\top$.

2.5.1 Worked applications

(11) Recovering ${}^C_I\mathbf{R}$ from vector pairs. Given camera frame $\{C\}$ and IMU frame $\{I\}$ with ${}^C\mathbf{v}_1 = {}^C_I\mathbf{R} {}^I\mathbf{v}_1$ and ${}^C\mathbf{v}_2 = {}^C_I\mathbf{R} {}^I\mathbf{v}_2$, the trick is to stack three independent directions (the third from the cross product) into an invertible matrix:

$$[{}^C\mathbf{v}_1 \ {}^C\mathbf{v}_2 \ {}^C\mathbf{v}_1 \times {}^C\mathbf{v}_2] = {}^C_I\mathbf{R} [{}^I\mathbf{v}_1 \ {}^I\mathbf{v}_2 \ {}^I\mathbf{v}_1 \times {}^I\mathbf{v}_2]. \quad (169)$$

(12) Given instead relative rotations ${}^{C_1}_{C_2}\mathbf{R} = {}^C_I\mathbf{R} {}^{I_1}_{I_2}\mathbf{R} {}^C_I\mathbf{R}^\top$, recover ${}^C_I\mathbf{R}$ from two such pairs.

(13) Discussion. For (11)/(12): What is the minimum number of data pairs needed? What requirements must the pairs satisfy? What if more than the minimum is available?

(14) Averaging noisy rotations. Given n measurements $\mathbf{R}_{m_i} = \mathbf{R} \exp([\mathbf{n}_i]_\times)$ with $\mathbf{n}_i \sim \mathcal{N}(\mathbf{0}_{3 \times 1}, \mathbf{P}_i)$, estimate $\mathbf{R}$ and its covariance.

2.6 Groups: SO(3) and SE(3)

SO(3) — special orthogonal group.

$$SO(3) = \{\mathbf{R} \in \mathbb{R}^{3 \times 3} \mid \mathbf{R}^\top \mathbf{R} = \mathbf{I}_3, \det(\mathbf{R}) = 1\}. \quad (170)$$

It satisfies the four group axioms: *closure* ($\mathbf{R}_1\mathbf{R}_2 \in SO(3)$), *identity* ($\mathbf{I}_3 \in SO(3)$), *inverse* ($\mathbf{R}^{-1} = \mathbf{R}^\top \in SO(3)$), and *associativity* ($(\mathbf{R}_1\mathbf{R}_2)\mathbf{R}_3 = \mathbf{R}_1(\mathbf{R}_2\mathbf{R}_3)$).

SE(3) — special Euclidean group.

$$SE(3) = \left\{ \mathbf{T} = \begin{bmatrix} \mathbf{R} & \mathbf{p} \\ \mathbf{0}_{1 \times 3} & 1 \end{bmatrix} \mid \mathbf{R} \in SO(3), \mathbf{p} \in \mathbb{R}^3 \right\}. \quad (171)$$

The four axioms: *closure*

$$\mathbf{T}_1\mathbf{T}_2 = \begin{bmatrix} \mathbf{R}_1 & \mathbf{p}_1 \\ \mathbf{0} & 1 \end{bmatrix} \begin{bmatrix} \mathbf{R}_2 & \mathbf{p}_2 \\ \mathbf{0} & 1 \end{bmatrix} \quad (172)$$

$$= \begin{bmatrix} \mathbf{R}_1\mathbf{R}_2 & \mathbf{R}_1\mathbf{p}_2 + \mathbf{p}_1 \\ \mathbf{0} & 1 \end{bmatrix} \in SE(3), \quad (173)$$

identity $\begin{bmatrix} \mathbf{I}_3 & \mathbf{0}_{3 \times 1} \\ \mathbf{0}_{1 \times 3} & 1 \end{bmatrix}$, *inverse*

$$\begin{bmatrix} \mathbf{R} & \mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix}^{-1} = \begin{bmatrix} \mathbf{R}^\top & -\mathbf{R}^\top \mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix} \in SE(3), \quad (174)$$

and *associativity* $(\mathbf{T}_1\mathbf{T}_2)\mathbf{T}_3 = \mathbf{T}_1(\mathbf{T}_2\mathbf{T}_3)$.

2.6.1 Matrix Lie groups

A *group* is a set with an operation satisfying closure, associativity, identity, and invertibility. A *Lie group* is a group that is also a smooth (differentiable) manifold whose group operation is smooth. A *matrix Lie group* has matrices as its elements. The maps between the vector space, the Lie algebra, and the Lie group are

$$\mathbb{R}^m \xrightarrow{\wedge \text{ (wedge)}} \mathfrak{so}(n) \xrightarrow{\exp} SO(n), \quad SO(n) \xrightarrow{\log} \mathfrak{so}(n) \xrightarrow{\vee \text{ (vee)}} \mathbb{R}^m, \quad (175)$$

and for short we write $\mathbf{X} = \text{Exp}(\mathbf{x}) = \exp(\mathbf{x}^\wedge)$ and $\mathbf{x} = \text{Log}(\mathbf{X}) = \log(\mathbf{X})^\vee$. For $SO(3)$ specifically, $\boldsymbol{\theta} \in \mathbb{R}^3 \rightarrow [\boldsymbol{\theta}]_\times = \boldsymbol{\theta}^\wedge \rightarrow \mathbf{R} = \exp(\boldsymbol{\theta}^\wedge)$, and $\mathbf{R} \in SO(3) \rightarrow [\boldsymbol{\theta}]_\times = \log(\mathbf{R}) \rightarrow \boldsymbol{\theta} = \log(\mathbf{R})^\vee$.

2.6.2 Why this (manifold) machinery?

It *simplifies* estimation: the core idea is to use a smooth manifold to model parameters that are awkward in plain vector space, which makes optimization, interpolation, integration, and differentiation easier or even possible (Fig. 3). For example, a raw angle $\theta \in [-\pi, \pi]$ has a wrap-around cut; the circle/sphere is smooth, and the tangent line at a point models the small change / error state used in optimization.

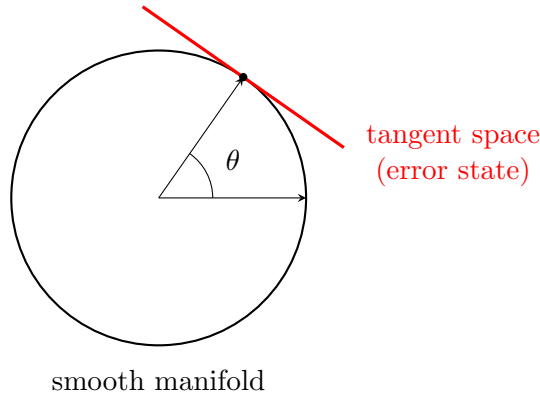


Figure 3: A smooth manifold (here a circle/sphere) avoids the wrap-around of a raw angle; the red tangent line is the local vector space used to model derivatives and error states for optimization.

For $SO(3)$ the left Jacobian also appears directly in the exponential expansion:

$$\mathbf{R}(\boldsymbol{\theta}) = \text{Exp}(\boldsymbol{\theta}) \quad (176)$$

$$= \mathbf{I}_3 + [\boldsymbol{\theta}]_\times \mathbf{J}_l(\boldsymbol{\theta}) = \mathbf{I}_3 + \boldsymbol{\theta}^\wedge \mathbf{J}_l(\boldsymbol{\theta}), \quad (177)$$

$$\mathbf{J}_l(\boldsymbol{\theta}) = \sum_{n=0}^{\infty} \frac{1}{(n+1)!} (\boldsymbol{\theta}^\wedge)^n. \quad (178)$$

2.7 SE(3)

Exp / Log. The $SE(3)$ tangent vector uses $\boldsymbol{\rho}$ for its translation component, $\boldsymbol{\xi} = \begin{bmatrix} \boldsymbol{\theta} \\ \boldsymbol{\rho} \end{bmatrix} \in \mathbb{R}^6$, which is *distinct* from the group translation $\mathbf{p}$ in $\mathbf{T} = \begin{bmatrix} \mathbf{R} & \mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix}$; the two are related by $\mathbf{p} = \mathbf{J}_l(\boldsymbol{\theta})\boldsymbol{\rho}$. With

the 4×4 algebra element $\boldsymbol{\xi}^\wedge = \begin{bmatrix} [\boldsymbol{\theta}]_\times & \boldsymbol{\rho} \\ \mathbf{0}_{1 \times 3} & 0 \end{bmatrix}$,

$$\mathbf{T} = \text{Exp}(\boldsymbol{\xi}) = \exp \begin{bmatrix} [\boldsymbol{\theta}]_\times & \boldsymbol{\rho} \\ \mathbf{0} & 0 \end{bmatrix} \quad (179)$$

$$= \begin{bmatrix} \exp([\boldsymbol{\theta}]_\times) & \mathbf{J}_l(\boldsymbol{\theta}) \boldsymbol{\rho} \\ \mathbf{0}_{1 \times 3} & 1 \end{bmatrix} \quad (180)$$

$$= \begin{bmatrix} \mathbf{R} & \mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix}, \quad \mathbf{p} = \mathbf{J}_l(\boldsymbol{\theta}) \boldsymbol{\rho}, \quad (181)$$

and inverting,

$$\boldsymbol{\xi} = \text{Log}(\mathbf{T})^\vee, \quad \log(\mathbf{T}) = \log \begin{bmatrix} \mathbf{R} & \mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix} \quad (182)$$

$$= \begin{bmatrix} \log(\mathbf{R}) & \mathbf{J}_l^{-1}(\boldsymbol{\theta}) \mathbf{p} \\ \mathbf{0} & 0 \end{bmatrix} \Rightarrow \boldsymbol{\xi} = \begin{bmatrix} \boldsymbol{\theta} \\ \mathbf{J}_l^{-1}(\boldsymbol{\theta}) \mathbf{p} \end{bmatrix} = \begin{bmatrix} \boldsymbol{\theta} \\ \boldsymbol{\rho} \end{bmatrix}. \quad (183)$$

Discussion. Prove $\exp \left(\begin{bmatrix} [\boldsymbol{\theta}]_\times & \boldsymbol{\rho} \\ \mathbf{0} & 0 \end{bmatrix} \right) = \begin{bmatrix} \mathbf{R} & \mathbf{J}_l(\boldsymbol{\theta}) \boldsymbol{\rho} \\ \mathbf{0} & 1 \end{bmatrix}$.

Left/right Jacobians. With $\text{Exp}(\boldsymbol{\xi} + \delta \boldsymbol{\xi}) = \text{Exp}(\boldsymbol{\xi}) \text{Exp}(\mathbf{J}_R(\boldsymbol{\xi}) \delta \boldsymbol{\xi}) = \text{Exp}(\mathbf{J}_L(\boldsymbol{\xi}) \delta \boldsymbol{\xi}) \text{Exp}(\boldsymbol{\xi})$,

$$\mathbf{J}_R(\boldsymbol{\xi}) = \begin{bmatrix} \mathbf{J}_R(\boldsymbol{\theta}) & \mathbf{0} \\ \mathbf{Q}_R(\boldsymbol{\xi}) & \mathbf{J}_R(\boldsymbol{\theta}) \end{bmatrix}_{6 \times 6}, \quad \mathbf{J}_L(\boldsymbol{\xi}) = \begin{bmatrix} \mathbf{J}_L(\boldsymbol{\theta}) & \mathbf{0} \\ \mathbf{Q}_L(\boldsymbol{\xi}) & \mathbf{J}_L(\boldsymbol{\theta}) \end{bmatrix}_{6 \times 6}, \quad (184)$$

and to first order, with $\boldsymbol{\xi}^\wedge = \begin{bmatrix} [\boldsymbol{\theta}]_\times & \mathbf{0} \\ [\boldsymbol{\rho}]_\times & [\boldsymbol{\theta}]_\times \end{bmatrix}_{6 \times 6}$,

$$\mathbf{J}_L(\boldsymbol{\xi}) \approx \mathbf{I}_6 + \frac{1}{2} \boldsymbol{\xi}^\wedge, \quad \mathbf{J}_R(\boldsymbol{\xi}) \approx \mathbf{I}_6 - \frac{1}{2} \boldsymbol{\xi}^\wedge. \quad (185)$$

Homework. Find $\mathbf{Q}_L$ (and $\mathbf{Q}_R$).

Error states: SE(3) vs SO(3) + $\mathbb{R}^3$. Perturb $\mathbf{T}$ on the right by the SE(3) tangent $\delta \boldsymbol{\xi} = [\delta \boldsymbol{\theta}^\top, \delta \boldsymbol{\rho}^\top]^\top$. Expanding with $\mathbf{J}_l(\delta \boldsymbol{\theta}) \approx \mathbf{I} + \frac{1}{2} [\delta \boldsymbol{\theta}]_\times$ and $\delta \mathbf{R} = \text{Exp}(\delta \boldsymbol{\theta})$,

$$\mathbf{T} \delta \mathbf{T} = \begin{bmatrix} \mathbf{R} & \mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix} \begin{bmatrix} \text{Exp}(\delta \boldsymbol{\theta}) & \mathbf{J}_l(\delta \boldsymbol{\theta}) \delta \boldsymbol{\rho} \\ \mathbf{0} & 1 \end{bmatrix} \quad (186)$$

$$= \begin{bmatrix} \mathbf{R} & \mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix} \begin{bmatrix} \delta \mathbf{R} & (\mathbf{I} + \frac{1}{2} [\delta \boldsymbol{\theta}]_\times) \delta \boldsymbol{\rho} \\ \mathbf{0} & 1 \end{bmatrix} \quad (187)$$

$$\approx \begin{bmatrix} \mathbf{R} & \mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix} \begin{bmatrix} \delta \mathbf{R} & \delta \boldsymbol{\rho} \\ \mathbf{0} & 1 \end{bmatrix} \quad (188)$$

$$= \begin{bmatrix} \mathbf{R} \delta \mathbf{R} & \mathbf{R} \delta \boldsymbol{\rho} + \mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix}. \quad (189)$$

If instead we treat the pose as $SO(3) + \mathbb{R}^3$ (rotation and translation perturbed separately, $\mathbf{p} \rightarrow \mathbf{p} + \delta\mathbf{p}$),

$$\mathbf{T} = \begin{bmatrix} \mathbf{R}\delta\mathbf{R} & \mathbf{p} + \delta\mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix} \quad (190)$$

$$= \begin{bmatrix} \mathbf{R}\delta\mathbf{R} & \mathbf{p} + \mathbf{R}\mathbf{R}^\top \delta\mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix} \quad (191)$$

$$= \begin{bmatrix} \mathbf{R} & \mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix} \begin{bmatrix} \delta\mathbf{R} & \mathbf{R}^\top \delta\mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix}. \quad (192)$$

Comparing the two conventions gives the relation between the error states ($SE(3)$ tangent translation $\delta\boldsymbol{\rho}$ vs. naive position error $\delta\mathbf{p}$),

$$\begin{bmatrix} \mathbf{R} & \mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix} \begin{bmatrix} \delta\mathbf{R} & \delta\mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix}_{SO(3)+\mathbb{R}^3} = \begin{bmatrix} \mathbf{R} & \mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix} \begin{bmatrix} \delta\mathbf{R} & \delta\boldsymbol{\rho} \\ \mathbf{0} & 1 \end{bmatrix}_{SE(3)} \Rightarrow \begin{cases} \delta\boldsymbol{\rho} = \mathbf{R}^\top \delta\mathbf{p} \\ \delta\mathbf{R}_{SE(3)} = \delta\mathbf{R} \end{cases}. \quad (193)$$

Adjoint. The adjoint transfers algebra elements across the group:

$$SO(3) : \mathbf{R} \text{Exp}(\boldsymbol{\varphi}) \mathbf{R}^\top = \text{Exp}(\mathbf{R}\boldsymbol{\varphi}) \Leftrightarrow \text{Ad}_{\mathbf{R}} = \mathbf{R}, \quad SE(3) : \mathbf{T} \text{Exp}(\boldsymbol{\xi}) \mathbf{T}^{-1} = \text{Exp}(\text{Ad}_{\mathbf{T}} \boldsymbol{\xi}), \quad (194)$$

with $\text{Ad}_{\mathbf{T}} = \begin{bmatrix} \mathbf{R} & \mathbf{0} \\ [\mathbf{p}]_{\times} \mathbf{R} & \mathbf{R} \end{bmatrix}_{6 \times 6}$. The adjoint is itself a Lie group, with $\text{Ad}_{\mathbf{T}} = \exp(\boldsymbol{\xi}^\wedge)$ and $\log(\text{Ad}_{\mathbf{T}})^\vee = \boldsymbol{\xi}$.

Homework. Prove $\mathbf{T} \text{Exp}(\boldsymbol{\xi}) \mathbf{T}^{-1} = \text{Exp}(\text{Ad}_{\mathbf{T}} \boldsymbol{\xi})$.

2.7.1 More operations

(1) Interpolation. For ${}^G\mathbf{T}_{t_1}, {}^G\mathbf{T}_{t_2}$, $t \in [t_1, t_2]$, $\lambda = \frac{t-t_1}{t_2-t_1}$: in $SE(3)$, with ${}^{t_1}\boldsymbol{\xi} = \text{Log}({}^G\mathbf{T}_{t_1}^{-1} {}^G\mathbf{T}_{t_2})$,

$${}^G_t\mathbf{T} = {}^G_{t_1}\mathbf{T} \text{Exp}(\lambda {}^{t_1}_{t_2}\boldsymbol{\xi}); \quad (195)$$

in $SO(3) + \mathbb{R}^3$, the rotation interpolates on the manifold and the position linearly:

$${}^G_t\mathbf{R} = {}^G_{t_1}\mathbf{R} \text{Exp}(\lambda \text{Log}({}^G_{t_1}\mathbf{R}^\top {}^G_{t_2}\mathbf{R})), \quad {}^G\mathbf{p}_t = {}^G\mathbf{p}_{t_1} + \lambda({}^G\mathbf{p}_{t_2} - {}^G\mathbf{p}_{t_1}). \quad (196)$$

(2) Time derivative. For $SE(3)$, with $\mathbf{T} = \mathbf{T} \text{Exp}([\delta\boldsymbol{\theta}; \delta\boldsymbol{\rho}]^\wedge)$ and body velocity, expand the limit using $\exp([\delta\boldsymbol{\theta}]_\times) - \mathbf{I}_3 \approx [\delta\boldsymbol{\theta}]_\times$ and $\mathbf{J}_l(\delta\boldsymbol{\theta}) \approx \mathbf{I} + \frac{1}{2}[\delta\boldsymbol{\theta}]_\times$:

$$\frac{d\mathbf{T}}{dt} = \lim_{\delta t \rightarrow 0} \frac{\mathbf{T}(t + \delta t) - \mathbf{T}(t)}{\delta t} \quad (197)$$

$$= \lim_{\delta t \rightarrow 0} \frac{1}{\delta t} \left(\begin{bmatrix} \mathbf{R} & \mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix} \begin{bmatrix} \exp([\delta\boldsymbol{\theta}]_\times) & \mathbf{J}_l(\delta\boldsymbol{\theta})\delta\boldsymbol{\rho} \\ \mathbf{0} & 1 \end{bmatrix} - \begin{bmatrix} \mathbf{R} & \mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix} \right) \quad (198)$$

$$= \lim_{\delta t \rightarrow 0} \frac{1}{\delta t} \begin{bmatrix} \mathbf{R} & \mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix} \begin{bmatrix} \exp([\delta\boldsymbol{\theta}]_\times) - \mathbf{I}_3 & \mathbf{J}_l(\delta\boldsymbol{\theta})\delta\boldsymbol{\rho} \\ \mathbf{0} & 0 \end{bmatrix} \quad (199)$$

$$= \lim_{\delta t \rightarrow 0} \frac{1}{\delta t} \begin{bmatrix} \mathbf{R} & \mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix} \begin{bmatrix} [\delta\boldsymbol{\theta}]_\times & \delta\boldsymbol{\rho} \\ \mathbf{0} & 0 \end{bmatrix} \quad (200)$$

$$= \begin{bmatrix} \mathbf{R} & \mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix} \begin{bmatrix} [\boldsymbol{\omega}]_\times & \mathbf{v} \\ \mathbf{0} & 0 \end{bmatrix} \quad (201)$$

$$= \begin{bmatrix} \mathbf{R}[\boldsymbol{\omega}]_\times & \mathbf{R}\mathbf{v} \\ \mathbf{0} & 0 \end{bmatrix}, \quad (202)$$

where $\mathbf{v}$ is the body velocity. For $SO(3) + \mathbb{R}^3$ the same computation with

$$\mathbf{T} = \begin{bmatrix} \mathbf{R} \exp([\delta\boldsymbol{\theta}]_\times) & \mathbf{p} + \delta\mathbf{p} \\ \mathbf{0} & 1 \end{bmatrix} \quad (203)$$

gives

$$\frac{d\mathbf{T}}{dt} = \begin{bmatrix} \mathbf{R}[\boldsymbol{\omega}]_\times & {}^G\mathbf{v} \\ \mathbf{0} & 0 \end{bmatrix}, \quad (204)$$

with ${}^G\mathbf{v}$ the global velocity.

(3) Calibration. For IMU frame $\{I\}$ and camera frame $\{C\}$,

$$\begin{matrix} C_1 \\ C_2 \end{matrix} \mathbf{T} = \begin{matrix} C \\ I \end{matrix} \mathbf{T} \begin{matrix} I_1 \\ I_2 \end{matrix} \mathbf{T} \begin{matrix} C \\ I \end{matrix} \mathbf{T}^{-1}, \quad \text{Exp}(\begin{matrix} C_1 \\ C_2 \end{matrix} \boldsymbol{\xi}) = \text{Exp}(\text{Ad}_{(\begin{matrix} C \\ I \end{matrix} \mathbf{T})} \begin{matrix} I_1 \\ I_2 \end{matrix} \boldsymbol{\xi}) \Rightarrow \begin{matrix} C_1 \\ C_2 \end{matrix} \boldsymbol{\xi} = \text{Ad}_{(\begin{matrix} C \\ I \end{matrix} \mathbf{T})} \begin{matrix} I_1 \\ I_2 \end{matrix} \boldsymbol{\xi}. \quad (205)$$

Homework. Compute the Jacobians $\partial_{C_2}^{C_1} \boldsymbol{\xi} / \partial_{I_2}^{I_1} \boldsymbol{\xi}$ and $\partial_{C_2}^{C_1} \boldsymbol{\xi} / \partial_I^C \boldsymbol{\xi}$ (and the corresponding $SO(3) + \mathbb{R}^3$ versions).

2.8 $SE_2(3)$

Augmenting the pose with a velocity column gives the 5×5 group element

$$\boldsymbol{\Upsilon} = \begin{bmatrix} \mathbf{R} & \mathbf{p} & \mathbf{v} \\ \mathbf{0}_{1 \times 3} & 1 & 0 \\ \mathbf{0}_{1 \times 3} & 0 & 1 \end{bmatrix}, \quad \boldsymbol{\xi} = \begin{bmatrix} \boldsymbol{\theta} \\ \boldsymbol{\rho} \\ \boldsymbol{\nu} \end{bmatrix} \rightarrow \boldsymbol{\xi}^\wedge = \begin{bmatrix} [\boldsymbol{\theta}]_\times & \boldsymbol{\rho} & \boldsymbol{\nu} \\ \mathbf{0} & 0 & 0 \\ \mathbf{0} & 0 & 0 \end{bmatrix}_{5 \times 5}, \quad (206)$$

where (as in $SE(3)$) the tangent components $\boldsymbol{\rho}, \boldsymbol{\nu}$ are distinct from the group position/velocity $\mathbf{p} = \mathbf{J}_l(\boldsymbol{\theta})\boldsymbol{\rho}$, $\mathbf{v} = \mathbf{J}_l(\boldsymbol{\theta})\boldsymbol{\nu}$. The exponential, logarithm, and adjoint follow the same pattern as $SE(3)$:

$$\text{Exp}(\boldsymbol{\xi}) = \begin{bmatrix} \mathbf{R} & \mathbf{J}_l(\boldsymbol{\theta})\boldsymbol{\rho} & \mathbf{J}_l(\boldsymbol{\theta})\boldsymbol{\nu} \\ \mathbf{0} & 1 & 0 \\ \mathbf{0} & 0 & 1 \end{bmatrix} = \begin{bmatrix} \mathbf{R} & \mathbf{p} & \mathbf{v} \\ \mathbf{0} & 1 & 0 \\ \mathbf{0} & 0 & 1 \end{bmatrix}, \quad (207)$$

$$\text{Log}(\boldsymbol{\Upsilon})^\vee = \begin{bmatrix} \boldsymbol{\theta} \\ \mathbf{J}_l^{-1}(\boldsymbol{\theta})\mathbf{p} \\ \mathbf{J}_l^{-1}(\boldsymbol{\theta})\mathbf{v} \end{bmatrix}, \quad \text{Ad}_{\boldsymbol{\Upsilon}} = \begin{bmatrix} \mathbf{R} & \mathbf{0} & \mathbf{0} \\ [\mathbf{p}]_\times \mathbf{R} & \mathbf{R} & \mathbf{0} \\ [\mathbf{v}]_\times \mathbf{R} & \mathbf{0} & \mathbf{R} \end{bmatrix}. \quad (208)$$

2.9 Quaternions

Quaternions are presented last because $SO(3)$, $SE(3)$, and the Lie-group view already cover most needs. There are two main conventions — *Hamilton* (right-handed) and *JPL* (left-handed); we use the **Hamilton** convention. For a thorough treatment see Solà [6] (Hamilton) and Trawny & Roumeliotis [5] (JPL).

A quaternion and its imaginary units satisfy

$$Q = q_w + q_x i + q_y j + q_z k, \quad i^2 = j^2 = k^2 = ijk = -1, \quad ij = k, \quad jk = i, \quad ki = j, \quad (209)$$

written as the 4×1 vector $\bar{q} = [q_w, q_x, q_y, q_z]^\top = [q_w, \mathbf{q}_v^\top]^\top$.

Product. The quaternion product is bilinear and can be written with left/right matrices:

$$\bar{p} \otimes \bar{q} = \mathcal{L}(\bar{p}) \bar{q} = \mathcal{R}(\bar{q}) \bar{p}, \quad (210)$$

$$\mathcal{L}(\bar{p}) = p_w \mathbf{I}_4 + \begin{bmatrix} 0 & -\mathbf{p}_v^\top \\ \mathbf{p}_v & [\mathbf{p}_v]_\times \end{bmatrix}, \quad (211)$$

$$\mathcal{R}(\bar{q}) = q_w \mathbf{I}_4 + \begin{bmatrix} 0 & -\mathbf{q}_v^\top \\ \mathbf{q}_v & -[\mathbf{q}_v]_\times \end{bmatrix}. \quad (212)$$

Identity, unit quaternion, inverse. The identity is $\bar{q} = [1, \mathbf{0}_{3 \times 1}^\top]^\top$. A rotation by angle θ about unit axis $\mathbf{k}$ is the unit quaternion

$$\bar{q} = \begin{bmatrix} \cos \frac{\theta}{2} \\ \sin \frac{\theta}{2} \mathbf{k} \end{bmatrix}, \quad \bar{q}^{-1} = \begin{bmatrix} \cos \frac{\theta}{2} \\ -\sin \frac{\theta}{2} \mathbf{k} \end{bmatrix}, \quad \mathbf{R}_{12} = \mathbf{R}_1 \mathbf{R}_2 \leftrightarrow \bar{q}_{12} = \bar{q}_1 \otimes \bar{q}_2. \quad (213)$$

Why $\theta/2$? — rotating a vector. With global frame $\{G\}$ and IMU frame $\{I\}$, rotate ${}^I \mathbf{p}$ via the sandwich product and convert to matrices with $\bar{p} \otimes \bar{q} = \mathcal{R}(\bar{q}) \bar{p}$ and $\bar{p} \otimes \bar{q} = \mathcal{L}(\bar{p}) \bar{q}$:

$$\begin{bmatrix} 0 \\ {}^G \mathbf{p} \end{bmatrix} = {}^G \bar{q} \otimes \begin{bmatrix} 0 \\ {}^I \mathbf{p} \end{bmatrix} \otimes {}^I \bar{q}^{-1} \quad (214)$$

$$= \mathcal{R}({}^G \bar{q}^{-1}) \mathcal{L}({}^G \bar{q}) \begin{bmatrix} 0 \\ {}^I \mathbf{p} \end{bmatrix}. \quad (215)$$

Writing $\bar{q} = [q_w, \mathbf{q}_v^\top]^\top$ and multiplying the two matrices,

$$\begin{aligned} \mathcal{R}(\bar{q}^{-1}) \mathcal{L}(\bar{q}) &= \left(q_w \mathbf{I}_4 + \begin{bmatrix} 0 & \mathbf{q}_v^\top \\ -\mathbf{q}_v & [\mathbf{q}_v]_\times \end{bmatrix} \right) \left(q_w \mathbf{I}_4 + \begin{bmatrix} 0 & -\mathbf{q}_v^\top \\ \mathbf{q}_v & [\mathbf{q}_v]_\times \end{bmatrix} \right) \\ &= q_w^2 \mathbf{I}_4 + 2q_w \begin{bmatrix} 0 & \mathbf{0} \\ \mathbf{0} & [\mathbf{q}_v]_\times \end{bmatrix} + \begin{bmatrix} \mathbf{q}_v^\top \mathbf{q}_v & \mathbf{0} \\ \mathbf{0} & -\mathbf{q}_v \mathbf{q}_v^\top + [\mathbf{q}_v]_\times^2 \end{bmatrix}, \end{aligned} \quad (216)$$

whose lower-right 3×3 block (acting on ${}^I \mathbf{p}$) gives

$${}^G \mathbf{p} = (q_w^2 \mathbf{I}_3 + 2q_w [\mathbf{q}_v]_\times + \mathbf{q}_v \mathbf{q}_v^\top + [\mathbf{q}_v]_\times^2) {}^I \mathbf{p}. \quad (217)$$

Using $[\mathbf{q}_v]_\times^2 = \mathbf{q}_v \mathbf{q}_v^\top - (\mathbf{q}_v^\top \mathbf{q}_v) \mathbf{I}_3$ and the unit-norm condition $q_w^2 + \mathbf{q}_v^\top \mathbf{q}_v = 1$,

$$q_w^2 \mathbf{I}_3 + \mathbf{q}_v \mathbf{q}_v^\top + [\mathbf{q}_v]_\times^2 = q_w^2 \mathbf{I}_3 + (\mathbf{q}_v^\top \mathbf{q}_v) \mathbf{I}_3 + 2 [\mathbf{q}_v]_\times^2 \quad (218)$$

$$= \mathbf{I}_3 + 2 [\mathbf{q}_v]_\times^2, \quad (219)$$

so

$${}^G\mathbf{p} = (\mathbf{I}_3 + 2q_w [\mathbf{q}_v]_{\times} + 2 [\mathbf{q}_v]_{\times}^2)^I \mathbf{p}. \quad (220)$$

Finally substitute $q_w = \cos \frac{\theta}{2}$, $\mathbf{q}_v = \sin \frac{\theta}{2} \mathbf{k}$, so $2q_w [\mathbf{q}_v]_{\times} = 2 \cos \frac{\theta}{2} \sin \frac{\theta}{2} [\mathbf{k}]_{\times} = \sin \theta [\mathbf{k}]_{\times}$ and $2 [\mathbf{q}_v]_{\times}^2 = 2 \sin^2 \frac{\theta}{2} [\mathbf{k}]_{\times}^2 = (1 - \cos \theta) [\mathbf{k}]_{\times}^2$:

$${}^G\mathbf{p} = (\mathbf{I}_3 + \sin \theta [\mathbf{k}]_{\times} + (1 - \cos \theta) [\mathbf{k}]_{\times}^2)^I \mathbf{p} \quad (221)$$

$$= {}^G_I \mathbf{R}^I \mathbf{p} \quad (222)$$

$$= \text{Exp}(\theta \mathbf{k})^I \mathbf{p}, \quad (223)$$

recovering Rodrigues' formula — which is why the half-angle $\theta/2$ appears.

Derivative. With $\delta \boldsymbol{\theta} = \delta \theta \mathbf{k}$ and small-angle $\delta \bar{q} = [\cos \frac{\delta \theta}{2}, \sin \frac{\delta \theta}{2} \mathbf{k}^{\top}]^{\top} \approx [1, \frac{1}{2} \delta \boldsymbol{\theta}^{\top}]^{\top}$,

$${}_t^G \dot{\bar{q}} = \lim_{\Delta t \rightarrow 0} \frac{{}_t^G \bar{q}_{t+\Delta t} - {}_t^G \bar{q}}{\Delta t} \quad (224)$$

$$= \lim_{\Delta t \rightarrow 0} \frac{{}_t^G \bar{q} \otimes \begin{bmatrix} \cos \frac{\delta \theta}{2} \\ \sin \frac{\delta \theta}{2} \mathbf{k} \end{bmatrix} - {}_t^G \bar{q} \otimes \begin{bmatrix} 1 \\ \mathbf{0} \end{bmatrix}}{\Delta t} \quad (225)$$

$$= \lim_{\Delta t \rightarrow 0} \frac{{}_t^G \bar{q} \otimes \begin{bmatrix} 0 \\ \frac{1}{2} \delta \boldsymbol{\theta} \end{bmatrix}}{\Delta t} \quad (226)$$

$$= {}_t^G \bar{q} \otimes \begin{bmatrix} 0 \\ \frac{1}{2} \boldsymbol{\omega} \end{bmatrix} \quad (227)$$

$$= \frac{1}{2} \mathcal{R} \left(\begin{bmatrix} 0 \\ \boldsymbol{\omega} \end{bmatrix} \right) {}_t^G \bar{q} \quad (228)$$

$$= \frac{1}{2} \begin{bmatrix} 0 & -\boldsymbol{\omega}^{\top} \\ \boldsymbol{\omega} & -[\boldsymbol{\omega}]_{\times} \end{bmatrix} {}_t^G \bar{q} \quad (229)$$

$$= \frac{1}{2} \Omega(\boldsymbol{\omega}) {}_t^G \bar{q}, \quad (230)$$

where $\Omega(\boldsymbol{\omega}) \triangleq \begin{bmatrix} 0 & -\boldsymbol{\omega}^{\top} \\ \boldsymbol{\omega} & -[\boldsymbol{\omega}]_{\times} \end{bmatrix}$.

Integration. Given ${}_t^G \bar{q}$, $\boldsymbol{\omega}$, δt , with $\delta \boldsymbol{\theta} = \boldsymbol{\omega} \delta t = \delta \theta \mathbf{k}$ and $\delta \bar{q} = [\cos \frac{\delta \theta}{2}, \sin \frac{\delta \theta}{2} \mathbf{k}^{\top}]^{\top}$,

$${}_{t+\delta t}^G \bar{q} = {}_t^G \bar{q} \otimes \delta \bar{q}. \quad (231)$$

3 Lecture 3: Camera Model, Triangulation, Point/Line/Plane

Topics. (1) camera measurements; (2) essential matrix / triangulation; (3) point / line / plane representations.

3.1 The camera measurement model

A camera produces pixel measurements that depend on the camera pose and the observed scene:

$$\mathbf{z}_c = \mathbf{h}_c(\mathbf{x}) + \mathbf{n}_c, \quad (232)$$

where $\mathbf{z}_c$ are the pixel measurements, $\mathbf{x}$ is the state vector (camera pose and point feature), $\mathbf{h}_c(\cdot)$ is the camera model, and $\mathbf{n}_c$ is the pixel noise. The geometry is shown in Fig. 4.

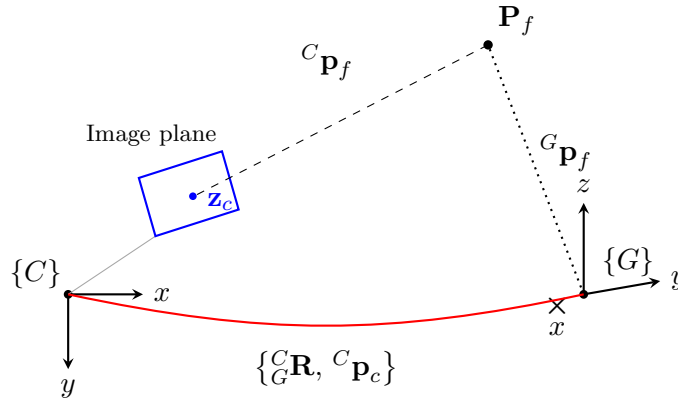


Figure 4: Camera measurement geometry: the feature $\mathbf{P}_f$ projects through the image plane to the pixel $\mathbf{z}_c$ in the camera frame $\{C\}$; the global frame $\{G\}$ and the extrinsic $\{^C_G \mathbf{R}, ^C \mathbf{p}_c\}$ relate the two views of the same point ($^C \mathbf{p}_f$ vs. $^G \mathbf{p}_f$).

Simplest pinhole model. With the feature in the camera frame $^C \mathbf{p}_f = [x, y, z]^\top$, the normalized projection is

$$\mathbf{z}_n = \begin{bmatrix} u_n \\ v_n \end{bmatrix} = \begin{bmatrix} x/z \\ y/z \end{bmatrix}, \quad \mathbf{z}_n = \mathbf{h}_p(^C \mathbf{p}_f) + \mathbf{n}_c, \quad ^C \mathbf{p}_f = ^C_G \mathbf{R}^\top (^G \mathbf{p}_f - ^G \mathbf{p}_c). \quad (233)$$

Linearizing the measurement, $\tilde{\mathbf{z}}_n = \mathbf{H}_c \tilde{\mathbf{x}} + \mathbf{n}_c$, with the residual

$$\tilde{\mathbf{z}}_n = \mathbf{z}_n - \mathbf{h}_p(^C \hat{\mathbf{p}}_f) = \begin{bmatrix} u_n \\ v_n \end{bmatrix} - \begin{bmatrix} \hat{x}/\hat{z} \\ \hat{y}/\hat{z} \end{bmatrix}. \quad (234)$$

The state and its error state are

$$\mathbf{x} : \{^C_G \mathbf{R}, ^G \mathbf{p}_c, ^G \mathbf{p}_f\}, \quad ^C_G \mathbf{R} = ^C_G \hat{\mathbf{R}} \text{Exp}(\delta \boldsymbol{\theta}), \quad ^G \mathbf{p}_c = ^G \hat{\mathbf{p}}_c + ^G \tilde{\mathbf{p}}_c, \quad ^G \mathbf{p}_f = ^G \hat{\mathbf{p}}_f + ^G \tilde{\mathbf{p}}_f, \quad (235)$$

with error state $\tilde{\mathbf{x}} = [\delta \boldsymbol{\theta}^\top, ^G \tilde{\mathbf{p}}_c^\top, ^G \tilde{\mathbf{p}}_f^\top]^\top$.

3.2 The measurement Jacobian

By the chain rule, factoring every block through ${}^C\tilde{\mathbf{p}}_f$,

$$\tilde{\mathbf{z}}_n = \mathbf{H}_c \tilde{\mathbf{x}} + \mathbf{n}_c \quad (236)$$

$$= \left[\frac{\partial \tilde{\mathbf{z}}_n}{\partial \delta \boldsymbol{\theta}} \quad \frac{\partial \tilde{\mathbf{z}}_n}{\partial {}^G\tilde{\mathbf{p}}_c} \quad \frac{\partial \tilde{\mathbf{z}}_n}{\partial {}^G\tilde{\mathbf{p}}_f} \right] \tilde{\mathbf{x}} + \mathbf{n}_c \quad (237)$$

$$= \frac{\partial \tilde{\mathbf{z}}_n}{\partial {}^C\tilde{\mathbf{p}}_f} \left[\frac{\partial {}^C\tilde{\mathbf{p}}_f}{\partial \delta \boldsymbol{\theta}} \quad \frac{\partial {}^C\tilde{\mathbf{p}}_f}{\partial {}^G\tilde{\mathbf{p}}_c} \quad \frac{\partial {}^C\tilde{\mathbf{p}}_f}{\partial {}^G\tilde{\mathbf{p}}_f} \right] \tilde{\mathbf{x}} + \mathbf{n}_c. \quad (238)$$

The projection Jacobian, writing ${}^C\mathbf{p}_f = [c_x, c_y, c_z]^\top$, is

$$\mathbf{z}_n = \begin{bmatrix} c_x/c_z \\ c_y/c_z \end{bmatrix}, \quad \frac{\partial \mathbf{z}_n}{\partial {}^C\mathbf{p}_f} = \begin{bmatrix} \frac{1}{c_z} & 0 & -\frac{c_x}{c_z^2} \\ 0 & \frac{1}{c_z} & -\frac{c_y}{c_z^2} \end{bmatrix} = \frac{1}{c_z^2} \begin{bmatrix} c_z & 0 & -c_x \\ 0 & c_z & -c_y \end{bmatrix}. \quad (239)$$

Perturbing ${}^C\mathbf{p}_f$. From ${}^C\mathbf{p}_f = {}^C\mathbf{R}^\top ({}^G\mathbf{p}_f - {}^G\mathbf{p}_c)$ with ${}^C\mathbf{R} = {}^C\hat{\mathbf{R}} \text{Exp}(\delta \boldsymbol{\theta})$ and $\text{Exp}(\delta \boldsymbol{\theta})^\top \approx \mathbf{I} - [\delta \boldsymbol{\theta}]_\times$,

$${}^C\hat{\mathbf{p}}_f + {}^C\tilde{\mathbf{p}}_f = ({}^C\hat{\mathbf{R}} \text{Exp}(\delta \boldsymbol{\theta}))^\top ({}^G\hat{\mathbf{p}}_f + {}^G\tilde{\mathbf{p}}_f - {}^G\hat{\mathbf{p}}_c - {}^G\tilde{\mathbf{p}}_c) \quad (240)$$

$$= (\mathbf{I} - [\delta \boldsymbol{\theta}]_\times) {}^C\hat{\mathbf{R}}^\top ({}^G\hat{\mathbf{p}}_f + {}^G\tilde{\mathbf{p}}_f - {}^G\hat{\mathbf{p}}_c - {}^G\tilde{\mathbf{p}}_c) \quad (241)$$

$$\approx {}^C\hat{\mathbf{R}}^\top ({}^G\hat{\mathbf{p}}_f - {}^G\hat{\mathbf{p}}_c) - [\delta \boldsymbol{\theta}]_\times {}^C\hat{\mathbf{R}}^\top ({}^G\hat{\mathbf{p}}_f - {}^G\hat{\mathbf{p}}_c) + {}^C\hat{\mathbf{R}}^\top ({}^G\tilde{\mathbf{p}}_f - {}^G\tilde{\mathbf{p}}_c). \quad (242)$$

Using $-[\delta \boldsymbol{\theta}]_\times \mathbf{a} = [\mathbf{a}]_\times \delta \boldsymbol{\theta}$ with $\mathbf{a} = {}^C\hat{\mathbf{p}}_f = {}^C\hat{\mathbf{R}}^\top ({}^G\hat{\mathbf{p}}_f - {}^G\hat{\mathbf{p}}_c)$ and matching hat/tilde terms,

$$\begin{aligned} {}^C\hat{\mathbf{p}}_f + {}^C\tilde{\mathbf{p}}_f &= \underbrace{{}^C\hat{\mathbf{R}}^\top ({}^G\hat{\mathbf{p}}_f - {}^G\hat{\mathbf{p}}_c)}_{{}^C\hat{\mathbf{p}}_f} + \left[{}^C\hat{\mathbf{R}}^\top ({}^G\hat{\mathbf{p}}_f - {}^G\hat{\mathbf{p}}_c) \right]_\times \delta \boldsymbol{\theta} \\ &\quad - {}^C\hat{\mathbf{R}}^\top {}^G\tilde{\mathbf{p}}_c + {}^C\hat{\mathbf{R}}^\top {}^G\tilde{\mathbf{p}}_f, \end{aligned} \quad (243)$$

so that, with ${}^C\hat{\mathbf{p}}_f = {}^C\hat{\mathbf{R}}^\top ({}^G\hat{\mathbf{p}}_f - {}^G\hat{\mathbf{p}}_c)$, the three blocks are

$$\frac{\partial {}^C\tilde{\mathbf{p}}_f}{\partial \delta \boldsymbol{\theta}} = [{}^C\hat{\mathbf{p}}_f]_\times, \quad \frac{\partial {}^C\tilde{\mathbf{p}}_f}{\partial {}^G\tilde{\mathbf{p}}_c} = -{}^C\hat{\mathbf{R}}^\top, \quad \frac{\partial {}^C\tilde{\mathbf{p}}_f}{\partial {}^G\tilde{\mathbf{p}}_f} = {}^C\hat{\mathbf{R}}^\top. \quad (244)$$

Therefore $\tilde{\mathbf{z}}_n = \mathbf{H}_c \tilde{\mathbf{x}} + \mathbf{n}_c$ with

$$\mathbf{H}_c = \frac{1}{c_z^2} \begin{bmatrix} c_z & 0 & -c_x \\ 0 & c_z & -c_y \end{bmatrix} \left[[{}^C\hat{\mathbf{p}}_f]_\times \quad -{}^C\hat{\mathbf{R}}^\top \quad {}^C\hat{\mathbf{R}}^\top \right]. \quad (245)$$

3.3 A few important discussions

(1) SLAM. The pose $\{{}^C\mathbf{R}, {}^G\mathbf{p}_c\}$ is the localization of the camera, and ${}^G\mathbf{p}_f$ is a 3D feature of the environment. Solving them together means we solve localization and mapping together $\Rightarrow$ *Simultaneous Localization And Mapping* (SLAM).

(2) **PnP / P3P — known feature, solve pose.** If ${}^G\mathbf{p}_f$ is known, given (u, v) , how do we solve $\{{}_G^C\mathbf{R}, {}_G^C\mathbf{p}_c\}$? Here $\mathbf{x} : \{{}_G^C\mathbf{R}, {}_G^C\mathbf{p}_c\}$ is 6-dof, while each measurement $\mathbf{z}_n = \mathbf{h}({}_G^C\mathbf{R}, {}_G^C\mathbf{p}_c) + \mathbf{n}_c$ contributes 2 dof, so we need at least 3 points ${}^C\mathbf{p}_{f_i} \leftrightarrow (u_i, v_i)$ to solve $\{{}_G^C\mathbf{R}, {}_G^C\mathbf{p}_c\}$.

- 3 pairs ${}^C\mathbf{p}_{f_i} \leftrightarrow (u_i, v_i)$, **P3P** (perspective-3-point), i.e. 2+1 points: use 3D $\leftrightarrow$ 3D correspondences to determine the same 3D pose.
- n pairs of 3D $\leftrightarrow$ 3D correspondences, **PnP** (perspective- n -point).
- **RANSAC** (random sample consensus) for outlier rejection.

This is the *structure* $\rightarrow$ *motion* direction (known structure, solve motion).

(3) **Triangulation — known poses, solve feature.** If $\{{}_G^C\mathbf{R}, {}_G^C\mathbf{p}_c\}$ is known, given (u_1, v_1) , (u_2, v_2) , solve $\mathbf{P}_f$ (Fig. 5). Each pixel gives a unit bearing and an unknown range r_i :

$$(u_1, v_1) \rightarrow {}^{C_1}\mathbf{b}_1 = \frac{[u_1, v_1, 1]^\top}{\|[u_1, v_1, 1]^\top\|}, \quad (u_2, v_2) \rightarrow {}^{C_2}\mathbf{b}_2 = \frac{[u_2, v_2, 1]^\top}{\|[u_2, v_2, 1]^\top\|}, \quad (246)$$

$${}^{C_1}\mathbf{p}_f = {}^{C_1}\mathbf{b}_1 r_1, \quad {}^{C_2}\mathbf{p}_f = {}^{C_2}\mathbf{b}_2 r_2, \quad (247)$$

with r_1, r_2 unknown. From ${}^{C_1}\mathbf{p}_f = {}_{C_2}^{C_1}\mathbf{R} {}^{C_2}\mathbf{p}_f + {}^{C_1}\mathbf{p}_{C_2}$ we obtain ${}^{C_1}\mathbf{b}_1 r_1 = {}_{C_2}^{C_1}\mathbf{R} {}^{C_2}\mathbf{b}_2 r_2 + {}^{C_1}\mathbf{p}_{C_2}$.

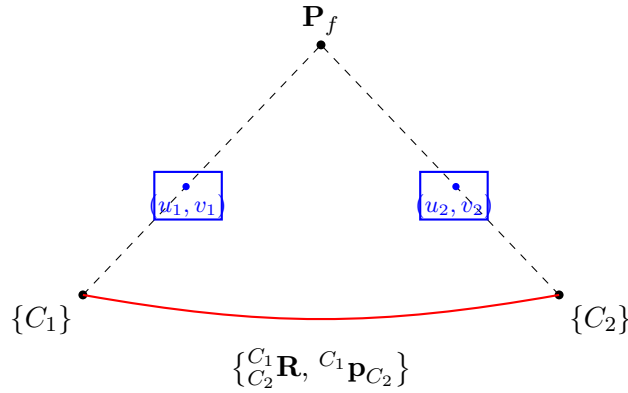


Figure 5: Two-view triangulation: bearings ${}^{C_1}\mathbf{b}_1, {}^{C_2}\mathbf{b}_2$ from pixels $(u_1, v_1), (u_2, v_2)$ and the known relative pose $\{{}_{C_2}^{C_1}\mathbf{R}, {}^{C_1}\mathbf{p}_{C_2}\}$ fix the feature $\mathbf{P}_f$ up to the two ranges r_1, r_2 .

Way 1 (stacked linear system). Collect the two ranges:

$${}^{C_1}\mathbf{b}_1 r_1 - {}_{C_2}^{C_1}\mathbf{R} {}^{C_2}\mathbf{b}_2 r_2 = {}^{C_1}\mathbf{p}_{C_2} \Rightarrow \underbrace{\begin{bmatrix} {}^{C_1}\mathbf{b}_1 & -{}_{C_2}^{C_1}\mathbf{R} {}^{C_2}\mathbf{b}_2 \end{bmatrix}}_{3 \times 2} \begin{bmatrix} r_1 \\ r_2 \end{bmatrix} = {}^{C_1}\mathbf{p}_{C_2}, \quad (248)$$

a 3×2 system; solving r_1, r_2 gives ${}^{C_1}\mathbf{p}_f, {}^{C_2}\mathbf{p}_f$.

Degenerate case. If the motion is pure rotation (no translation), then ${}^{C_1}\mathbf{b}_1 = {}_{C_2}^{C_1}\mathbf{R} {}^{C_2}\mathbf{b}_2$ and ${}^{C_1}\mathbf{p}_{C_2} = \mathbf{0}_{3 \times 1}$, so $r_1 = r_2$ and there is no way to solve r . For a monocular camera, pure rotation is dangerous.

Way 2 (left null space). With ${}^{C_1}\mathbf{b}_1 r_1 = {}_{C_2}^{C_1}\mathbf{R} {}^{C_2}\mathbf{b}_2 r_2 + {}^{C_1}\mathbf{p}_{C_2}$, collect into $\mathbf{B}^\perp = [\mathbf{b}^{\perp_1} \quad \mathbf{b}^{\perp_2}]$ two directions orthogonal to ${}^{C_1}\mathbf{b}_1$ (${}^{C_1}\mathbf{b}_1^\top \mathbf{b}^{\perp_1} = 0$, ${}^{C_1}\mathbf{b}_1^\top \mathbf{b}^{\perp_2} = 0$); projecting removes r_1 :

$$\mathbf{B}^{\perp \top} {}^{C_1}\mathbf{b}_1 r_1 = \mathbf{B}^{\perp \top} ({}_{C_2}^{C_1}\mathbf{R} {}^{C_2}\mathbf{b}_2 r_2 + {}^{C_1}\mathbf{p}_{C_2}) = \mathbf{0}_{2 \times 1} \Rightarrow \underbrace{\mathbf{B}^{\perp \top} {}_{C_2}^{C_1}\mathbf{R} {}^{C_2}\mathbf{b}_2}_{2 \times 1} r_2 = - \underbrace{\mathbf{B}^{\perp \top} {}^{C_1}\mathbf{p}_{C_2}}_{2 \times 1}. \quad (249)$$

(4) **Monocular triangulation is up to scale.** If $\{^C_G \mathbf{R}\}$ is known but for $^{C_1} \mathbf{p}_{C_2}$ we only know the direction $\overrightarrow{^{C_1} \mathbf{p}_{C_2}}$ (with $\|^{C_1} \mathbf{p}_{C_2}\|$ unknown), can we triangulate $^C \mathbf{p}_f$? Let $\|^{C_1} \mathbf{p}_{C_2}\| = \rho$, then $^{C_1} \mathbf{p}_f = \frac{C_1}{C_2} \mathbf{R} \frac{C_2}{C_1} \mathbf{p}_f + ^{C_1} \mathbf{p}_{C_2}$ gives

$$^{C_1} \mathbf{b}_1 r_1 = \frac{C_1}{C_2} \mathbf{R} \frac{C_2}{C_1} \mathbf{b}_2 r_2 + \rho \overrightarrow{^{C_1} \mathbf{p}_{C_2}}. \quad (250)$$

With $\mathbf{A} = [\mathbf{v}^{\perp_1} \ \mathbf{v}^{\perp_2}]^\top$ built from $\overrightarrow{^{C_1} \mathbf{p}_{C_2}}^\top \mathbf{v}^{\perp_1} = 0$, $\overrightarrow{^{C_1} \mathbf{p}_{C_2}}^\top \mathbf{v}^{\perp_2} = 0$, projecting kills the translation term:

$$\mathbf{A} \frac{C_1}{C_2} \mathbf{b}_1 r_1 = \mathbf{A} \frac{C_1}{C_2} \mathbf{R} \frac{C_2}{C_1} \mathbf{b}_2 r_2 + \rho \underbrace{\mathbf{A} \overrightarrow{^{C_1} \mathbf{p}_{C_2}}}_0 \quad (251)$$

$$\Rightarrow \begin{bmatrix} \mathbf{A} \frac{C_1}{C_2} \mathbf{b}_1 & -\mathbf{A} \frac{C_1}{C_2} \mathbf{R} \frac{C_2}{C_1} \mathbf{b}_2 \end{bmatrix} \begin{bmatrix} r_1 \\ r_2 \end{bmatrix} = 0. \quad (252)$$

Hence r_1, r_2 are up to scale: a monocular camera does not have scale. The above uses *motion to infer structure*.

(5) **Essential matrix — solve the motion.** Given $\{u_1, v_1\}, \{u_2, v_2\}$, can we solve $\{^C_G \mathbf{R}, ^C \mathbf{p}_c\}$? The three vectors $^{C_1} \mathbf{p}_f$, $^{C_2} \mathbf{p}_f$, $^{C_1} \mathbf{p}_{C_2}$ are coplanar (Fig. 6), so $^{C_1} \mathbf{p}_{C_2} \times ^{C_1} \mathbf{p}_f \perp \frac{C_1}{C_2} \mathbf{R} \frac{C_2}{C_1} \mathbf{p}_f$:

$$(\lfloor ^{C_1} \mathbf{p}_{C_2} \rfloor_\times \frac{C_1}{C_2} \mathbf{p}_f)^\top \frac{C_1}{C_2} \mathbf{R} \frac{C_2}{C_1} \mathbf{p}_f = 0 \Rightarrow ^{C_1} \mathbf{p}_f^\top \lfloor ^{C_1} \mathbf{p}_{C_2} \rfloor_\times \frac{C_1}{C_2} \mathbf{R} \frac{C_2}{C_1} \mathbf{p}_f = 0. \quad (253)$$

With $^{C_1} \mathbf{p}_f = ^{C_1} \mathbf{b}_1 r_1$, $^{C_2} \mathbf{p}_f = ^{C_2} \mathbf{b}_2 r_2$,

$$\boxed{^{C_1} \mathbf{b}_1^\top \lfloor ^{C_1} \mathbf{p}_{C_2} \rfloor_\times \frac{C_1}{C_2} \mathbf{R} \frac{C_2}{C_1} \mathbf{b}_2 = 0, \quad \mathbf{E} = \lfloor ^{C_1} \mathbf{p}_{C_2} \rfloor_\times \frac{C_1}{C_2} \mathbf{R} \quad (\text{essential matrix}).} \quad (254)$$

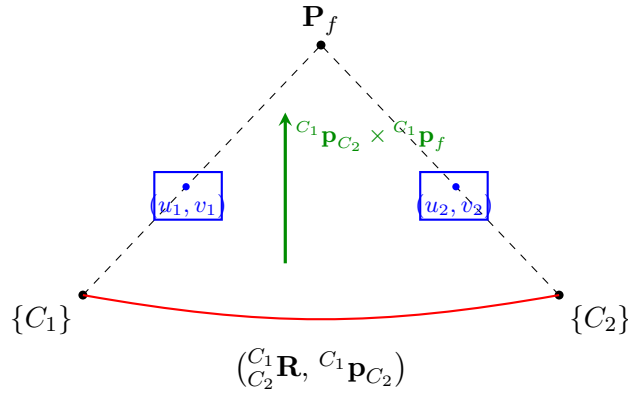


Figure 6: Epipolar geometry: $^{C_1} \mathbf{p}_f$, $^{C_2} \mathbf{p}_f$, and the baseline $^{C_1} \mathbf{p}_{C_2}$ are coplanar, giving the essential-matrix constraint $^{C_1} \mathbf{b}_1^\top \mathbf{E} ^{C_2} \mathbf{b}_2 = 0$.

Why is $\mathbf{E}$ 5-dof? Each pair $(u_1, v_1), (u_2, v_2)$ provides 1 constraint, so we need at least **5 pairs of points** to solve $\mathbf{E} \Rightarrow \frac{C_1}{C_2} \mathbf{R}$ (3 dof) and $\overrightarrow{^{C_1} \mathbf{p}_{C_2}}$ (2 dof, direction only). When selecting pairs of points, RANSAC (random sample consensus) comes in. Finally:

$$\text{normalized coordinates: } ^{C_1} \mathbf{b}_1^\top \lfloor ^{C_1} \mathbf{p}_{C_2} \rfloor_\times \frac{C_1}{C_2} \mathbf{R} \frac{C_2}{C_1} \mathbf{b}_2 = 0 \quad (\text{essential matrix}), \quad (255)$$

$$\text{pixel coordinates: fundamental matrix.} \quad (256)$$

This is the *motion from structure* direction.

3.4 The full intrinsic calibration model

The calibration geometry is the same as Fig. 4 (repeated here as Fig. 7). The camera model is a composition of four functions.

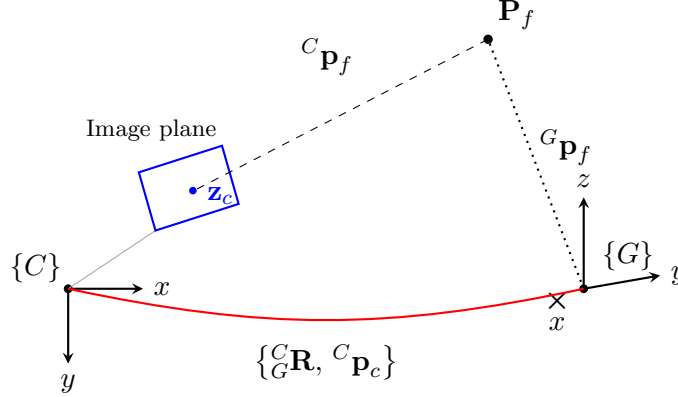


Figure 7: The intrinsic calibration sits inside the same projection geometry: the 3D feature is transformed into $\{C\}$, projected, distorted, and mapped to pixels.

Projection and transform.

$$\mathbf{z}_n = \begin{bmatrix} c_x/c_z \\ c_y/c_z \end{bmatrix} \triangleq \mathbf{h}_p({}^C \mathbf{p}_f), \quad {}^C \mathbf{p}_f = {}^C \mathbf{R}^\top ({}^G \mathbf{p}_f - {}^G \mathbf{p}_c) \triangleq \mathbf{h}_t({}^C \mathbf{R}, {}^G \mathbf{p}_c, {}^G \mathbf{p}_f), \quad (257)$$

$$\mathbf{z}_n = \mathbf{h}_p(\mathbf{h}_t({}^C \mathbf{R}, {}^G \mathbf{p}_c, {}^G \mathbf{p}_f)) + \mathbf{n}. \quad (258)$$

Distortion. Radial-tangential (for example); fisheye uses the equidistant model. With $r^2 = u_n^2 + v_n^2$ and distortion parameters $\mathbf{x}_d = [k_1, k_2, p_1, p_2]^\top$,

$$\begin{bmatrix} u_d \\ v_d \end{bmatrix} = \begin{bmatrix} u_n \\ v_n \end{bmatrix} (1 + k_1 r^2 + k_2 r^4) + \begin{bmatrix} 2p_1 u_n v_n + p_2 (r^2 + 2u_n^2) \\ p_1 (r^2 + 2v_n^2) + 2p_2 u_n v_n \end{bmatrix}, \quad \mathbf{z}_d = \begin{bmatrix} u_d \\ v_d \end{bmatrix} = \mathbf{h}_d(\mathbf{z}_n). \quad (259)$$

Intrinsics / homography. With $\mathbf{x}_{in} = [f_u, f_v, c_u, c_v]$,

$$\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \underbrace{\begin{bmatrix} f_u & 0 & c_u \\ 0 & f_v & c_v \\ 0 & 0 & 1 \end{bmatrix}}_{\mathbf{K}} \begin{bmatrix} u_d \\ v_d \\ 1 \end{bmatrix}, \quad \mathbf{z} = \begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} f_u & 0 \\ 0 & f_v \end{bmatrix} \begin{bmatrix} u_d \\ v_d \end{bmatrix} + \begin{bmatrix} c_u \\ c_v \end{bmatrix} \triangleq \mathbf{h}_k(\mathbf{z}_d, \mathbf{x}_{in}). \quad (260)$$

Full composition.

$$\mathbf{z}_c = \mathbf{h}_c(\mathbf{x}) + \mathbf{n}_c = \mathbf{h}_k(\mathbf{z}_d, \mathbf{x}_{in}) + \mathbf{n}_c \quad (261)$$

$$= \mathbf{h}_k(\mathbf{h}_d(\mathbf{z}_n, \mathbf{x}_d), \mathbf{x}_{in}) + \mathbf{n}_c \quad (262)$$

$$= \mathbf{h}_k(\mathbf{h}_d(\mathbf{h}_p({}^C \mathbf{p}_f), \mathbf{x}_d), \mathbf{x}_{in}) + \mathbf{n}_c \quad (263)$$

$$= \mathbf{h}_k(\mathbf{h}_d(\mathbf{h}_p(\mathbf{h}_t({}^C \mathbf{R}, {}^G \mathbf{p}_c, {}^G \mathbf{p}_f)), \mathbf{x}_d), \mathbf{x}_{in}) + \mathbf{n}_c, \quad (264)$$

where $\mathbf{h}_k(\cdot)$ is the camera-intrinsics homography, $\mathbf{h}_d(\cdot)$ the distortion function, $\mathbf{h}_p(\cdot)$ the projection function, $\mathbf{h}_t(\cdot)$ the transform function, $\mathbf{x}_d$ the distortion parameters, and $\mathbf{x}_{in} = \{f_u, f_v, c_u, c_v\}$ the intrinsics. The Jacobians for calibration follow by the chain rule.

3.5 The extrinsic (camera–IMU) calibration

The camera and IMU sensor frames are rigidly connected (Fig. 8).

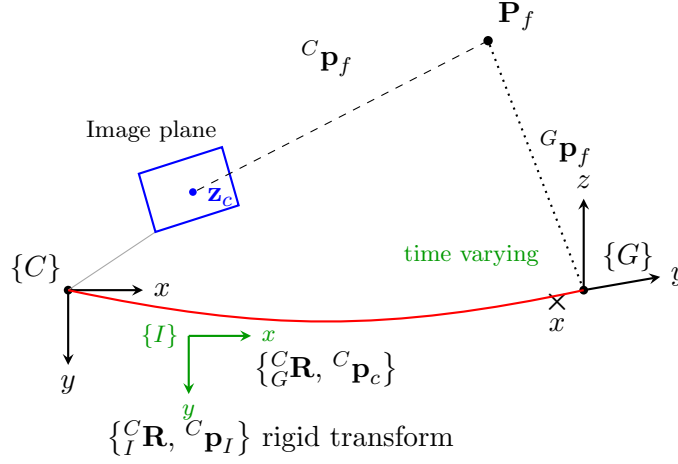


Figure 8: Extrinsic calibration: the IMU frame $\{I\}$ is rigidly fixed to the camera $\{C\}$ via $\{^C_I \mathbf{R}, ^C_I \mathbf{p}_I\}$, while the camera-to-global transform is time varying.

$$\begin{bmatrix} ^C \mathbf{p}_f \\ 1 \end{bmatrix} = \underbrace{\begin{bmatrix} ^C_I \mathbf{R} & ^C_I \mathbf{p}_I \\ \mathbf{0}_{1 \times 3} & 1 \end{bmatrix}}_{^C_I \mathbf{T}} \underbrace{\begin{bmatrix} ^G_I \mathbf{R} & ^G_I \mathbf{p}_I \\ \mathbf{0}_{1 \times 3} & 1 \end{bmatrix}^{-1}}_{^G_I \mathbf{T}^{-1}} \begin{bmatrix} ^G \mathbf{p}_f \\ 1 \end{bmatrix}, \quad (265)$$

$$^C \mathbf{p}_f = ^C_I \mathbf{R} ^I \mathbf{p}_f + ^C_I \mathbf{p}_I = ^C_I \mathbf{R} (^G_I \mathbf{R} (^G \mathbf{p}_f - ^G_I \mathbf{p}_I)) + ^C_I \mathbf{p}_I. \quad (266)$$

Time offset. How do we estimate the time offset between camera and IMU? With $t_I = t_c + t_d$,

$$\begin{bmatrix} ^{C_{t_c}} \mathbf{p}_f \\ 1 \end{bmatrix} = ^C_I \mathbf{T} \cdot (^{G_{t_c+t_d}} \mathbf{T})^{-1} \begin{bmatrix} ^G \mathbf{p}_f \\ 1 \end{bmatrix}, \quad (267)$$

$$^{G_{t_c+t_d}} \dot{\mathbf{T}} = \begin{bmatrix} ^G_I \mathbf{R} & ^G_I \mathbf{p}_I \\ \mathbf{0}_{1 \times 3} & 1 \end{bmatrix} \begin{bmatrix} [^I \boldsymbol{\omega}]_{\times} & ^I \mathbf{v} \\ \mathbf{0}_{1 \times 3} & 0 \end{bmatrix}. \quad (268)$$

which is the Jacobian for t_d . This is then the full calibration for the vision system.

3.6 Point / line / plane representation

3.6.1 Point representation

In $\mathbb{R}^3$ (closest point),

$$\mathbf{P}_f = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = z \begin{bmatrix} x/z \\ y/z \\ 1 \end{bmatrix} = z \begin{bmatrix} u_n \\ v_n \\ 1 \end{bmatrix}. \quad (269)$$

Inverse depth. Store the inverse of the depth, $\lambda \triangleq 1/z$:

$$\mathbf{x}_f = \begin{bmatrix} x \\ y \\ 1/z \end{bmatrix} = \begin{bmatrix} x \\ y \\ \lambda \end{bmatrix}. \quad (270)$$

What is the motivation to use inverse depth? Transform the feature into a second view and divide by z :

$${}^{C_2}\mathbf{p}_f = {}^{C_2}_{C_1}\mathbf{R}({}^{C_1}\mathbf{p}_f - {}^{C_1}\mathbf{p}_{C_2}) = {}^{C_2}_{C_1}\mathbf{R}\left(\begin{bmatrix} x \\ y \\ z \end{bmatrix} - {}^{C_1}\mathbf{p}_{C_2}\right) = {}^{C_2}_{C_1}\mathbf{R}\left(z \begin{bmatrix} x/z \\ y/z \\ 1 \end{bmatrix} - {}^{C_1}\mathbf{p}_{C_2}\right), \quad (271)$$

$$\frac{{}^{C_2}\mathbf{p}_f}{z} = {}^{C_2}_{C_1}\mathbf{R}\left(\begin{bmatrix} x/z \\ y/z \\ 1 \end{bmatrix} - \frac{1}{z}{}^{C_1}\mathbf{p}_{C_2}\right) \Rightarrow \begin{bmatrix} {}^{C_2}x/z \\ {}^{C_2}y/z \\ {}^{C_2}z/z \end{bmatrix} = {}^{C_2}_{C_1}\mathbf{R}\left(\begin{bmatrix} x\lambda \\ y\lambda \\ 1 \end{bmatrix} - \lambda {}^{C_1}\mathbf{p}_{C_2}\right). \quad (272)$$

If $z \rightarrow \infty$, the feature is far away, $z \approx {}^{C_2}z \gg {}^{C_2}x, {}^{C_2}y$, then the left side tends to the bearing measurement in $\{C_2\}$:

$$\begin{bmatrix} {}^{C_2}x/z \\ {}^{C_2}y/z \\ {}^{C_2}z/z \end{bmatrix} \approx \underbrace{\begin{bmatrix} u_{n_2} \\ v_{n_2} \\ 1 \end{bmatrix}}_{\text{measurements in } C_2} \propto {}^{C_2}_{C_1}\mathbf{R}\left(\begin{bmatrix} x\lambda \\ y\lambda \\ 1 \end{bmatrix} - \lambda {}^{C_1}\mathbf{p}_{C_2}\right). \quad (273)$$

The Jacobian with respect to the inverse-depth state $\tilde{\mathbf{x}} = [\tilde{x}, \tilde{y}, \tilde{\lambda}]$ is

$$\frac{\partial[u_{n_2}; v_{n_2}]}{\partial \tilde{\mathbf{x}}} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} [\mathbf{H}_\theta \quad \mathbf{H}_p \quad \mathbf{H}_{p_f}], \quad (274)$$

with the three blocks

$$\mathbf{H}_\theta = \left[{}^{C_2}_{C_1}\mathbf{R}([x\lambda; y\lambda; 1] - \lambda {}^{C_1}\mathbf{p}_{C_2}) \right]_{\times}, \quad \mathbf{H}_p = \underbrace{-\lambda {}^{C_2}_{C_1}\mathbf{R}}_{\lambda \rightarrow 0}, \quad (275)$$

$$\mathbf{H}_{p_f} = {}^{C_2}_{C_1}\mathbf{R} \left[\underbrace{\lambda \mathbf{e}_1}_0 \quad \underbrace{\lambda \mathbf{e}_2}_0 \quad [x; y; 0] - {}^{C_1}\mathbf{p}_{C_2} \right]. \quad (276)$$

If the feature is far away, then the rotation is improved, but the Jacobian for translation will not be improved (the $-\lambda {}^{C_2}_{C_1}\mathbf{R}$ and $\lambda \mathbf{e}_1, \lambda \mathbf{e}_2$ columns vanish as $\lambda \rightarrow 0$).

3.6.2 Plane representation

A plane is parameterized by its normal direction and distance to the origin (Fig. 9a):

$$\boldsymbol{\pi} = \begin{bmatrix} \mathbf{n}_\pi \\ d_\pi \end{bmatrix}_{4 \times 1}, \quad \mathbf{n}_\pi : \text{normal direction}, \quad d_\pi : \text{distance to origin}. \quad (277)$$

A plane is 3-dof but $\boldsymbol{\pi}$ has 4 parameters. The *closest-point representation* packs both into one vector,

$$\mathbf{P}_\pi = d_\pi \mathbf{n}_\pi = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \quad (278)$$

the closest point from the plane to the origin.



Figure 9: Plane representation (a) and its transfer between the camera and IMU frames (b).

Transfer $\{C\} \rightarrow \{I\}$. For ${}^C\mathbf{p}_\pi \leftrightarrow {}^I\mathbf{p}_\pi$ (Fig. 9b),

$$\begin{cases} {}^I\mathbf{n}_\pi = {}^I_C\mathbf{R} {}^C\mathbf{n}_\pi & \Rightarrow \text{normal direction} \\ {}^I d_\pi = {}^C d_\pi - {}^C\mathbf{n}_\pi^\top \cdot {}^C\mathbf{p}_I & \Rightarrow \text{origin to plane} \end{cases} \quad (279)$$

Hence, using ${}^I d_\pi = {}^C d_\pi - {}^C\mathbf{p}_I^\top {}^C\mathbf{n}_\pi$ and ${}^C\mathbf{p}_I = -{}^C_I\mathbf{R} {}^I\mathbf{p}_C$ (so $-{}^C\mathbf{p}_I^\top = {}^I\mathbf{p}_C^\top {}^I_C\mathbf{R}$),

$$\begin{bmatrix} {}^I\mathbf{n}_\pi \\ {}^I d_\pi \end{bmatrix} = \begin{bmatrix} {}^I_C\mathbf{R} & \mathbf{0}_{3 \times 1} \\ -{}^C\mathbf{p}_I^\top & 1 \end{bmatrix} \begin{bmatrix} {}^C\mathbf{n}_\pi \\ {}^C d_\pi \end{bmatrix} = \begin{bmatrix} {}^I_C\mathbf{R} & \mathbf{0}_{3 \times 1} \\ {}^I\mathbf{p}_C^\top {}^I_C\mathbf{R} & 1 \end{bmatrix} \begin{bmatrix} {}^C\mathbf{n}_\pi \\ {}^C d_\pi \end{bmatrix}. \quad (280)$$

The transformation chain is therefore

$${}^G\mathbf{p}_\pi \rightarrow ({}^C\mathbf{n}_\pi, {}^C d_\pi) \rightarrow ({}^I_C\mathbf{R}, {}^I\mathbf{p}_C) \rightarrow ({}^I\mathbf{n}_\pi, {}^I d_\pi) \rightarrow {}^I\mathbf{p}_\pi. \quad (281)$$

3.6.3 Line representation

How do we represent a 3D line (Fig. 10)? For a line L : $\mathbf{v}_e$ is the 3D direction, d_L is the distance from the origin to the line, and $\mathbf{n}_e$ is the normal direction of the plane (through the line and the origin).

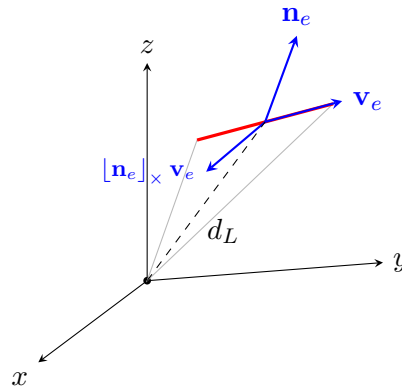


Figure 10: 3D line L : direction $\mathbf{v}_e$, plane normal $\mathbf{n}_e$, third axis $[\mathbf{n}_e]_\times \mathbf{v}_e$, and perpendicular distance d_L from the origin.

We can represent a line with $\{\mathbf{n}_e, \mathbf{v}_e, d_L\}$, where

$$\mathbf{R}_L = \begin{bmatrix} \mathbf{n}_e & \mathbf{v}_e & [\mathbf{n}_e]_\times \mathbf{v}_e \end{bmatrix} \text{ (a rotation matrix),} \quad d_L \text{ (a distance scalar).} \quad (282)$$

So $L : \{\mathbf{R}_L, d_L\} : \{\bar{\mathbf{q}}_L, d_L\}$, packed as the 4D vector $\mathbf{P}_L = \bar{\mathbf{q}}_L \cdot d_L$, with the error states in Euclidean space. Then

$$\mathbf{P}_L = \hat{\mathbf{P}}_L + \tilde{\mathbf{P}}_L = \bar{\mathbf{q}}_L \cdot d_L = \hat{\mathbf{q}}_L \otimes \delta \mathbf{q}_L \cdot (\hat{d}_L + \tilde{d}_L) \quad (283)$$

$$\Leftrightarrow \hat{\mathbf{P}}_L + \tilde{\mathbf{P}}_L = \hat{\mathbf{q}}_L \otimes \begin{bmatrix} 1 \\ \frac{1}{2} \delta \boldsymbol{\theta}_L \end{bmatrix} (\hat{d}_L + \tilde{d}_L) \quad (284)$$

$$= \hat{\mathbf{q}}_L \otimes \begin{bmatrix} \hat{d}_L + \tilde{d}_L \\ \frac{1}{2} \hat{d}_L \delta \boldsymbol{\theta}_L \end{bmatrix} \quad (285)$$

$$= \hat{\mathbf{q}}_L \otimes \left(\begin{bmatrix} \hat{d}_L \\ \mathbf{0}_{3 \times 1} \end{bmatrix} + \begin{bmatrix} 1 & \mathbf{0} \\ \mathbf{0} & \frac{1}{2} \hat{d}_L \mathbf{I}_3 \end{bmatrix} \begin{bmatrix} \tilde{d}_L \\ \delta \boldsymbol{\theta}_L \end{bmatrix} \right), \quad (286)$$

so that, with $[\hat{\mathbf{q}}_L]_L$ the left quaternion-product matrix,

$$\tilde{\mathbf{P}}_L = [\hat{\mathbf{q}}_L]_L \begin{bmatrix} 1 & \mathbf{0} \\ \mathbf{0} & \frac{1}{2} \hat{d}_L \mathbf{I}_3 \end{bmatrix} \begin{bmatrix} \tilde{d}_L \\ \delta \boldsymbol{\theta}_L \end{bmatrix}. \quad (287)$$

A few discussions on lines with cameras. (1) How to project a 3D line to a 2D line in the image? Given a line $\{\mathbf{n}_e, \mathbf{v}_e, d_L\}$ and camera parameters f_u, f_v, c_u, c_v (Fig. 11a):

$$\begin{bmatrix} u_n \\ v_n \\ 1 \end{bmatrix} \sim \underbrace{\begin{bmatrix} f_u & 0 & c_u \\ 0 & f_v & c_v \\ 0 & 0 & 1 \end{bmatrix}}_{\mathbf{K}} \underbrace{\begin{bmatrix} x \\ y \\ z \end{bmatrix}}_{\mathbf{p}_f}, \quad \underbrace{\mathbf{p}}_{\text{pixel on image}} \sim \mathbf{K} \mathbf{p}_f. \quad (288)$$

Every $\mathbf{p}_f$ on the line lies in the plane through the line and the origin, so $\mathbf{p}_f^\top \mathbf{n}_e = 0$. With $\mathbf{p}_f \sim \mathbf{K}^{-1} \mathbf{p}$,

$$(\mathbf{K}^{-1} \mathbf{p})^\top \mathbf{n}_e = 0 \Rightarrow \mathbf{p}^\top \mathbf{K}^{-\top} \mathbf{n}_e = 0 \Rightarrow \mathbf{n}_e^\top \mathbf{K}^{-1} \mathbf{p} = 0 \Rightarrow (\mathbf{K}^{-\top} \mathbf{n}_e)^\top \mathbf{p} = 0, \quad (289)$$

so the 2D image line is

$$\mathbf{l} = \mathbf{K}^{-\top} \mathbf{n}_e, \quad \mathbf{p} \text{ is every pixel on the line.} \quad (290)$$

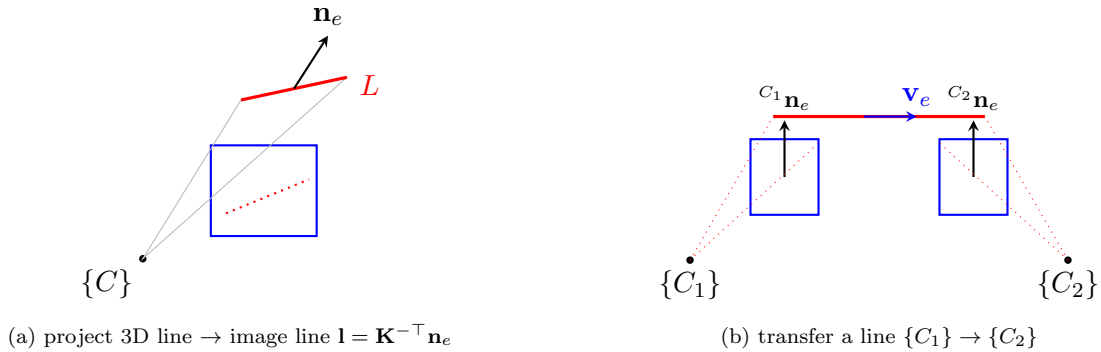


Figure 11: Lines with cameras: projecting a 3D line to a 2D image line (a) and transferring a line between two camera frames (b).

(2) How to transfer a line from frame $\{C_1\}$ to $\{C_2\}$? (Fig. 11b) The line L is given by ${}^{C_1}\mathbf{n}_e$, ${}^{C_1}\mathbf{v}_e$, ${}^{C_1}d_L$ in $\{C_1\}$ and by ${}^{C_2}\mathbf{n}_e$, ${}^{C_2}\mathbf{v}_e$, ${}^{C_2}d_L$ in $\{C_2\}$. Assume ${}^{C_1}\mathbf{p}_f$ is a point on L ; then its moment about the origin is

$$\left[{}^{C_1}\mathbf{p}_f \right]_{\times} {}^{C_1}\mathbf{v}_e = {}^{C_1}\mathbf{n}_e {}^{C_1}d_L. \quad (291)$$

The direction transforms by rotation only, and the point by the full rigid transform:

$${}^{C_2}\mathbf{v}_e = {}^{C_2}\mathbf{R} {}^{C_1}\mathbf{v}_e, \quad {}^{C_2}\mathbf{p}_f = {}^{C_2}\mathbf{R} {}^{C_1}\mathbf{p}_f + {}^{C_2}\mathbf{p}_{C_1}. \quad (292)$$

Hence the moment in $\{C_2\}$ is

$$\left[{}^{C_2}\mathbf{p}_f \right]_{\times} {}^{C_2}\mathbf{v}_e = \left[{}^{C_2}\mathbf{R} {}^{C_1}\mathbf{p}_f + {}^{C_2}\mathbf{p}_{C_1} \right]_{\times} {}^{C_2}\mathbf{R} {}^{C_1}\mathbf{v}_e = {}^{C_2}\mathbf{n}_e {}^{C_2}d_L. \quad (293)$$

Using $[\mathbf{R}\mathbf{a}]_{\times} \mathbf{R} = \mathbf{R} [\mathbf{a}]_{\times}$ on the first term,

$${}^{C_2}\mathbf{R} \left[{}^{C_1}\mathbf{p}_f \right]_{\times} {}^{C_1}\mathbf{v}_e + \left[{}^{C_2}\mathbf{p}_{C_1} \right]_{\times} {}^{C_2}\mathbf{R} {}^{C_1}\mathbf{v}_e = {}^{C_2}\mathbf{n}_e {}^{C_2}d_L, \quad (294)$$

and substituting the moment identity ${}^{C_1}\mathbf{n}_e {}^{C_1}d_L = \left[{}^{C_1}\mathbf{p}_f \right]_{\times} {}^{C_1}\mathbf{v}_e$,

$$\left[{}^{C_2}\mathbf{R} \left[{}^{C_2}\mathbf{p}_{C_1} \right]_{\times} {}^{C_2}\mathbf{R} \right] \begin{bmatrix} {}^{C_1}\mathbf{n}_e {}^{C_1}d_L \\ {}^{C_1}\mathbf{v}_e \end{bmatrix} = {}^{C_2}\mathbf{n}_e {}^{C_2}d_L. \quad (295)$$

Stacking the moment and the direction (${}^{C_2}\mathbf{v}_e = {}^{C_2}\mathbf{R} {}^{C_1}\mathbf{v}_e$) gives the full line transfer:

$$\begin{bmatrix} {}^{C_2}\mathbf{n}_e {}^{C_2}d_L \\ {}^{C_2}\mathbf{v}_e \end{bmatrix} = \begin{bmatrix} {}^{C_2}\mathbf{R} & \left[{}^{C_2}\mathbf{p}_{C_1} \right]_{\times} {}^{C_2}\mathbf{R} \\ \mathbf{0}_3 & {}^{C_2}\mathbf{R} \end{bmatrix} \begin{bmatrix} {}^{C_1}\mathbf{n}_e {}^{C_1}d_L \\ {}^{C_1}\mathbf{v}_e \end{bmatrix}. \quad (296)$$

References

- [1] Timothy D. Barfoot. *State Estimation for Robotics*. See Chapter 7 (matrix Lie groups). Cambridge University Press, 2017.
- [2] Frank Dellaert and GTSAM contributors. *Lie Groups for 2D and 3D Transformations*. GT-SAM documentation, borglab/gtsam. URL: <https://github.com/borglab/gtsam/blob/develop/doc/LieGroups.pdf>.
- [3] Nikolay Atanasov. *Rotations $SO(3)$ and Rigid-Body Motions $SE(3)$* . ECE276A: Sensing & Estimation in Robotics, Lecture 12, UC San Diego. 2020. URL: https://natanaso.github.io/ece276a2020/ref/ECE276A_12_S03_SE3.pdf.
- [4] Joan Solà, Jérémie Deray, and Dinesh Atchuthan. “A Micro Lie Theory for State Estimation in Robotics”. In: *arXiv preprint arXiv:1812.01537* (2018). URL: <https://arxiv.org/pdf/1812.01537>.
- [5] Nikolas Trawny and Stergios I. Roumeliotis. *Indirect Kalman Filter for 3D Attitude Estimation*. Tech. rep. 2005-002. See Section 1 (quaternion algebra). University of Minnesota, Dept. of Computer Science & Engineering, MARS Lab, 2005. URL: https://mediawiki.isr.tecnico.ulisboa.pt/images/d/db/Indirect_Kalman_Filter_for_3D_Attitude_Estimation.pdf.
- [6] Joan Solà. “Quaternion Kinematics for the Error-State Kalman Filter”. In: *arXiv preprint arXiv:1711.02508* (2017). See Sections 1–3 (quaternion definition, properties, conversions). URL: <https://www.iri.upc.edu/people/jsola/JoanSola/objectes/notes/kinematics.pdf>.